

A concrete extension principle for fuzzy set theory

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ABSTRACT

In Fuzzy Set Theory (FST), Zadeh's Extension Principle (ZEP) has been a precise method for extending mathematical functions from classical crisp sets to fuzzy sets. However, some of the key components of ZEP are difficult to interpret, especially for users without deep FST expertise. Additionally, applying ZEP to fundamental tasks, such as extending classical arithmetic to fuzzy settings, can produce results too imprecise for practical applications. This article proposes an alternative FST extension principle with several advantages: (i) *Well-Defined Procedure*: It offers a clear method for extending classical mathematical functions to their fuzzy counterparts. (ii) *Concrete and Comprehensible Ingredients*: The principle consists of elements understandable to FST users, particularly non-experts. This ensures users not only receive a solution but also understand the steps leading to it. (iii) *Adaptive Fuzzy Functions*: The proposed principle allows users to construct fuzzy functions that adapt to their problem domains. (iv) *Versatility in Application*: To demonstrate its versatility as a *unified* approach to solving non-trivial tasks, we apply the principle to (a) calculate the probabilities of fuzzy events and (b) illustrate how to derive user-interpretable fuzzy-valued statistical measures systematically.

1. Introduction

Since Zadeh's seminal work in 1965 (Zadeh, 1965), fuzzy set theory (FST) has been extensively studied both theoretically and in practical applications. FST has been applied to various fields, including engineering, medicine, and management. Theoretical advancements in FST have been equally diverse. For example, researchers have developed type-2 fuzzy sets (e.g., Mendel, 2014; Mendel & John, 2002; Mendel, Liu, & Zhai, 2009; Mizumoto & Tanaka, 1981; Nieminen, 1977) and other higher-order fuzzy sets (e.g., Atanassov, Vassilev, & Tsvetkov, 2013; Ngan, 2016, 2017a, 2017b; Rodriguez, Martinez, Torra, Xu, & Herrera, 2014). Additionally, significant progress has been made in extending set-theoretic and logical operators for fuzzy sets (e.g., Gera & Dombi, 2008; Hernandez, Cubillo, & Torres-Blanc, 2015; Walker & Walker, 2009; Wang & Hu, 2015) and advancing fuzzy arithmetic (e.g., Boukezzoula, Galichet, Foulloy, & Elmasry, 2014; Chalco-Cano, Lodwick, & Bede, 2014; Hanss, 2002; Ngan, 2021).

Following Zadeh (1965), a fuzzy set is defined as follows:

Definition 1.1. (Fuzzy sets) Given a universe of discourse U , a fuzzy set A in U is characterized by a membership function $\mu_A : U \rightarrow [0, 1]$. For each element $u \in U$, $\mu_A(u)$ provides a real value from the interval $[0, 1]$, expressing the degree of membership of u in A . Notation-wise, the fuzzy set A can be represented as $A = \{ \langle u, \mu_A(u) \rangle \mid u \in U \}$. **End of Definition.**

Underneath the theoretical and practical advances in FST lies a central concept: the extension principle. Developed by Zadeh (1965, 1975), Zadeh's extension principle (ZEP) has served as a fundamental approach for extending mathematical functions for classical crisp sets to the corresponding functions for fuzzy sets. Formally, ZEP can be described as follows:

Definition 1.2. (Zadeh's extension principle) Let U be the Cartesian product of the universes of discourse $U = U_1 \times \dots \times U_N$. Suppose f is a mapping from U to another universe of discourse Y , such that for any given $(u_1, \dots, u_N) \in U$, there exists a unique $y \in Y$ with $y = f(u_1, \dots, u_N)$. Given N fuzzy sets $A_1, \dots, A_N$ of $U_1, \dots, U_N$ respectively, Zadeh's extension principle states that the fuzzy version $\hat{f}$ of the function f is defined as:

$$C = \hat{f}(A_1, \dots, A_N) \quad (1.1)$$

where C is a fuzzy set of Y , with

$$C \equiv \{ \langle y, \mu_C(y) \rangle \mid y \in Y \} \quad (1.2)$$

and

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$$\mu_C(y) = \begin{cases} \sup_{(u_1, \dots, u_N) \in f^{-1}(y)} \{\min\{\mu_{A_1}(u_1), \dots, \mu_{A_N}(u_N)\}\} & \text{when } f^{-1}(y) \text{ is not empty} \\ 0 & \text{otherwise} \end{cases} \quad (1.3)$$

End of Definition.

With the above definition, ZEP offers a syntactically precise procedure for transforming a classical function into its fuzzy counterpart. However, its key operations and constructs can be challenging to interpret, especially for FST users without deep FST expertise. Syntactically, identifying C involves: (i) for each y in Y , using f^{-1} to find all relevant elements $(u_1, \dots, u_N)$ (Eqs. (1.2) and (1.3)), and (ii) executing the “supremum among the minimums” operation to obtain the membership value for each y (Eq. (1.3)). While this is a computationally well-defined procedure to obtain a unique membership value for each y in Y , an FST user may find it challenging to justify the appropriateness of using the supremum among the minimums in their application area. For example, in the context of common-sense reasoning in a court trial, let C represent the concept of a *Highly Plausible Conclusion*. Suppose that a legal team gathers five separate pieces of strong evidence supporting a particular hypothesis y^* , corresponding to five distinct solutions to the equation:

$$y^* = f(u_1, \dots, u_N) \quad (1.4)$$

leading to the following collection of values:

$$\{\min\{\mu_{A_1}(u_1), \dots, \mu_{A_N}(u_N)\} | (u_1, \dots, u_N) \in f^{-1}(y^*)\} = \{0.9, 0.9, 0.9, 0.9, 0.9\} \quad (1.5)$$

On the other hand, suppose that the opposing legal team presents only a single piece of strong evidence supporting an alternative hypothesis y^{**} , corresponding to a single solution to the equation

$$y^{**} = f(u_1, \dots, u_N) \quad (1.6)$$

leading to:

$$\{\min\{\mu_{A_1}(u_1), \dots, \mu_{A_N}(u_N)\} | (u_1, \dots, u_N) \in f^{-1}(y^{**})\} = \{0.9\} \quad (1.7)$$

In a *supremum-based* extension principle like Definition 1.2, combining Eq. (1.3) with Eq. (1.5) and with Eq. (1.7) yield $\mu_C(y^*) = 0.9$ and $\mu_C(y^{**}) = 0.9$ respectively. This indicates that both y^* and y^{**} are considered to have the same degree of belongingness to the fuzzy set C of *Highly Plausible Conclusion*. However, one could argue that Eq. (1.5) provides more substantial evidence supporting the belongingness of y^* to C than Eq. (1.7) does for y^{**} , given the collection of five 0.9's versus just one 0.9. This suggests an alternative view that $\mu_C(y^*) > \mu_C(y^{**})$. For many FST users without deep FST expertise, this alternative might seem more reasonable, at least in some situations, based on common-sense reasoning. Therefore, it is desirable for an extension principle to accommodate such possibilities. (In Section 4, we will further address this issue.)

Additionally, directly applying ZEP to fundamental tasks, such as extending classical arithmetic to the FST setting, can lead to results that are too imprecise for real-life applications (e.g., see Chalco-Cano et al., 2014; G. J. Klir, 1997; Piegat, 2005). Efforts to modify ZEP have been made, but as we will discuss in Section 4, these modifications have primarily occurred at the syntactic level. The new constructs introduced in the modified extension principles, along with the old constructs these modified principles have retained, arguably still suffer from being non-trivial to interpret, especially for the FST users who are non-experts.

To make FST more broadly applicable to real-life domains, it is crucial to address the issue of user-interpretability. In this article, we aim to tackle this problem by developing an FST extension principle that is as concrete and user-interpretable as feasible, specifically for cases where each of $U_1, \dots, U_N$ mentioned in Definition 1.2 is $\mathfrak{N}$. The proposed FST extension principle is based on a K -intervals representation of fuzzy

numbers described in Ngan (2021) and an extension principle for interval analysis described in Ngan (2024). The contributions of this article include:

- (i) *Syntactically Well-Defined Procedure*: The proposed extension principle offers a clear method for extending classical mathematical functions to their fuzzy counterparts.
- (ii) *Concrete and Comprehensible Constructs*: The principle is composed of operations and constructs that are understandable to FST users, such as classically trained engineers who may not possess deep FST expertise. This ensures that users not only obtain a solution to their problem but also understand the entire sequence of steps leading to the solution.
- (iii) *Adaptive Fuzzy Functions*: The proposed extension principle allows FST users to construct fuzzy functions that are adaptive to the assumptions in their problem domains.
- (iv) *Versatility in Application*: To demonstrate its versatility as a *unified* approach to solving various non-trivial tasks, we will apply the proposed extension principle to (a) calculate the probabilities of fuzzy events in a user-interpretable manner, and to (b) illustrate how user-interpretable fuzzy-valued statistical measures can be derived systematically.

The rest of this article is organized as follows: In Sections 2.1 and 2.2, we revisit the K -intervals representation of fuzzy numbers discussed in Ngan (2021) and the theoretical results concerning the extension principle for interval analysis discussed in Ngan (2024). In the remainder of Section 2, we develop our proposed extension principle for FST based on these methods. In Section 3, we apply the proposed extension principle to solve problems involving the probabilities of fuzzy events and illustrate the development of user-interpretable fuzzy-valued statistical measures. Section 4 compares our approach with previous efforts in the FST literature. Finally, in the Conclusion section, we discuss the limitations of the present approach and outline future research directions.

2. Development of a user-interpretable FST extension principle

As described in the Introduction, the development of our proposed extension principle is based upon the K -intervals representation for fuzzy numbers (Ngan, 2021) and the extension principle for interval analysis (Ngan, 2024). We summarize these methods in Sections 2.1 and 2.2. Then, in Section 2.3, we develop an FST extension principle based on these methods.

2.1. A K -intervals representation of fuzzy numbers

Building upon Definition 1.1, the class of fuzzy numbers is a subclass of fuzzy sets of the real line $\mathfrak{N}$ obeying the following property:

Definition 2.1. (*Fuzzy numbers* (G. Klir & Yuan, 1995)) Given the universe of discourse $\mathfrak{N}$ and a fuzzy set A of $\mathfrak{N}$, A is termed a fuzzy number if its membership function μ_A is such that:

$$\mu_A(x) = \begin{cases} \mu_A^L(x) & \text{when } a \leq x \leq b \\ 1 & \text{when } b \leq x \leq c \\ \mu_A^R(x) & \text{when } c \leq x \leq d \\ 0 & \text{otherwise} \end{cases} \quad (2.1)$$

where the terms a , b , c and d are given real values such that $a < b \leq c < d$, $\mu_A^L(x) : [a, b] \rightarrow [0, 1]$ is continuous and monotonic increasing, and $\mu_A^R(x) : [c, d] \rightarrow [0, 1]$ is continuous and monotonic decreasing. **End of Definition.**

Building upon Definition 2.1, Ngan (2021) defines the K -intervals representation of a fuzzy number as follows:

Definition 2.2. (The K -intervals representation of a fuzzy number) Given a fuzzy number A as defined according to Eq. (2.1), we:

- (i) Denote the inverses of $\mu_A^L(x)$ and $\mu_A^R(x)$ as $\nu_A^L(\alpha) : [0, 1] \rightarrow [a, b]$ and $\nu_A^R(\alpha) : [0, 1] \rightarrow [c, d]$ respectively.
- (ii) Define the 'flip' function $\mu_{A,F}^L$ of μ_A^L :

$$\mu_{A,F}^L(x) \equiv \begin{cases} 1 - \mu_A^L(x) & \text{if } a \leq x \leq b \\ 0 & \text{if } b < x \leq (b+c)/2 \end{cases} \quad (2.2)$$

and the 'un-flip' function $\mu_{A,U}^R$ of μ_A^R :

$$\mu_{A,U}^R(x) \equiv \begin{cases} \mu_A^R(x) & \text{if } c \leq x \leq d \\ 1 & \text{if } (b+c)/2 \leq x < c \end{cases} \quad (2.3)$$

- (iii) Denote, for $a \leq x \leq b$, the inverse of $\mu_{A,F}^L(x)$ as $\nu_{A,F}^L(\alpha) : [0, 1] \rightarrow [a, b]$, and for $c \leq x \leq d$, the inverse of $\mu_{A,U}^R(x)$ as $\nu_{A,U}^R(\alpha) : [0, 1] \rightarrow [c, d]$.
- (iv) Subdivide the membership interval $[0, 1]$ into K levels, with $\alpha_k \equiv (k - 1/2)/K$ and $k \in \{1, \dots, K\}$, and finally define $A^k \equiv [x_{A^k}^L, x_{A^k}^R]$, where $x_{A^k}^L = \nu_{A,F}^L(\alpha_k)$ and $x_{A^k}^R = \nu_{A,U}^R(\alpha_k)$.

We denote the resultant set of intervals $S_A \equiv \{A^k | k \in \{1, \dots, K\}\}$ as the K -intervals representation of the fuzzy number A . **End of Definition.**

To illustrate Definition 2.2, consider the fuzzy number depicted in Fig. 1a. By 'flipping' μ_A^L but not 'flipping' μ_A^R , the graphs of $\mu_{A,F}^L$ and $\mu_{A,U}^R$ are obtained and illustrated in Fig. 1b. Furthermore, in Fig. 1b, the membership interval $[0, 1]$ is subdivided into K levels, and we associate $A^k \equiv [x_{A^k}^L, x_{A^k}^R]$ as the interval chosen by a Decision Maker (DM) k . (Hence, we have a total of K DMs.) Suppose the fuzzy number represents the concept "worthy of a B + grade". Then, for a very stringent DM, a score in the interval $[x_{A^1}^L, x_{A^1}^R] = [73, 80]$ in Fig. 1b would be considered "B + worthy", whereas for a very lenient DM, a score in the interval $[x_{A^K}^L, x_{A^K}^R] = [68, 75]$ would be considered "B + worthy".

Consider three possible cases:

- (i) If we are given a crisp value x such that $a \leq x \leq b$ (say $x = 71$), from Fig. 1b, we see that the percentage of the K DMs whose intervals contain this x equals $1 - \mu_{A,F}^L(x)$. Note that this quantity is equal to $\mu_A^L(x)$ (due to Eq. (2.2)) and hence to $\mu_A(x)$ (due to Eq. (2.1)).
- (ii) If we are given an x such that $c \leq x \leq d$ (say $x = 76.5$), we see that the percentage of the K DMs whose intervals contain this x equals $\mu_{A,U}^R(x) - 0$. This quantity is equal to $\mu_A^R(x)$ (due to Eq. (2.3)) and hence to $\mu_A(x)$ (due to Eq. (2.1)).
- (iii) Finally, if we are given an x such that $b \leq x \leq c$ (say $x = 74$), we see that the corresponding percentage equals $1 - 0 = 1$ (as this x is contained in every DM's interval, which is equal to $\mu_A(x)$ (due to Eq. (2.1)).

Therefore, in all cases (i), (ii) and (iii), under the K -intervals representation, the percentage of all intervals that contain a given crisp value x is equal to the membership value $\mu_A(x)$. Let us encapsulate the above observation as follows:

Proposition 2.1. (Extracting membership values from the K -intervals representation) Given a fuzzy number A and its membership function μ_A as defined according to Eq. (2.1), we obtain its K -intervals representation $S_A \equiv \{A^k : k \in \{1, \dots, K\}\}$ according to Definition 2.2. Then, for any $x \in \mathfrak{N}$, the following holds:

$$\mu_A(x) = \lim_{K \rightarrow \infty} \sum_{k=1}^K Q(x, A^k) / K \quad (2.4.1)$$

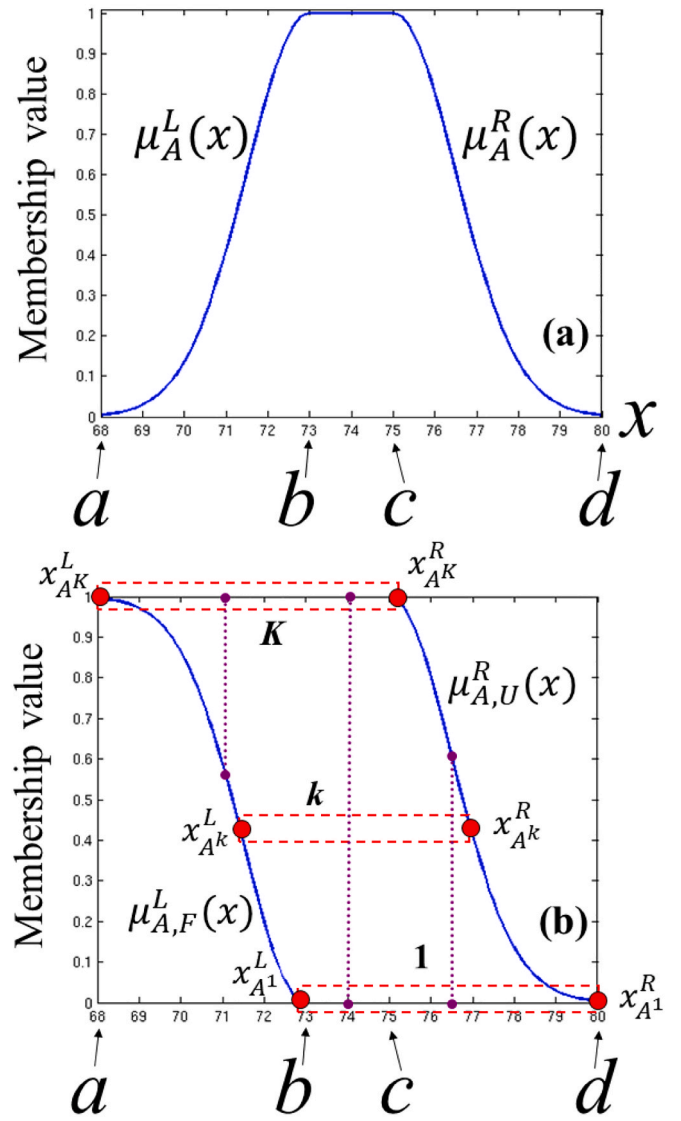


Fig. 1. Representation of a fuzzy number in terms of its K numerical intervals. (a) A given fuzzy number. (b) The K -intervals representation of the fuzzy number.

$$\text{where } Q(x, I) \equiv \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{otherwise} \end{cases} \text{ for any interval } I \text{ in } \mathfrak{N} \quad (2.4.2)$$

End of Proposition.

In summary: (a) The concept of a fuzzy number in the FST setting can be concretely represented as a set of K numerical intervals in the classical mathematics setting, each interval being associated with a DM. (b) The notion of the membership degree of an element in the FST setting can be classically expressed as the percentage of the DMs whose intervals contain that element.

In the rest of this article, any reference to $Q(\bullet, \bullet)$ refers to the indicator function defined in Eq. (2.4.2).

2.1.1. Advantages of the K -intervals representation of fuzzy numbers

In the FST literature, the membership function of a fuzzy number is often divided into a collection of intervals termed α -cuts (e.g., see Hanks, 2005; Mendel, 2017; Zimmermann, 2001), as shown in Fig. 2a. We argue that the K -intervals representation of fuzzy numbers has at least two advantages over the α -cuts representation:

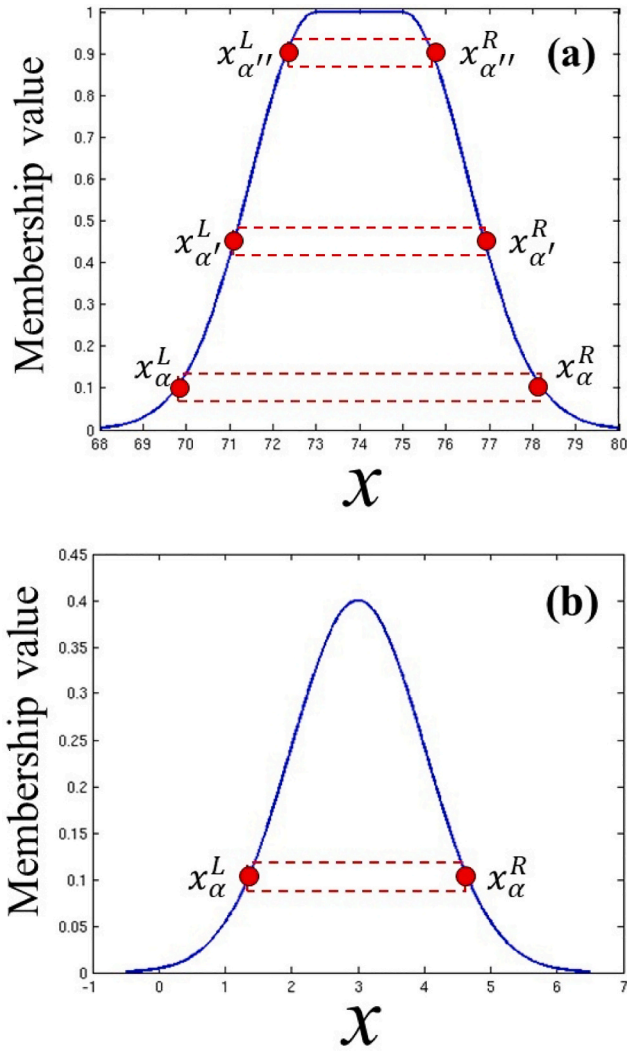


Fig. 2. The α -cuts representation of a fuzzy number. (a) The given fuzzy number is normal (i.e., the maximum membership value equals 1). The α -cuts progressively shorten as the α -value increases from α , to α' , and to α'' . (b) The given fuzzy number is subnormal. The α -cut becomes an empty set when the α -value is greater than 0.4.

- (i) *Equal Weight Treatment*: As illustrated in Fig. 1b, the intervals assigned to the DMs have comparable lengths, making it reasonable to treat the K intervals (and hence the K DMs) equally, i.e., with equal weights. This “equal weight” treatment naturally leads to the concrete and intuitive interpretation that the membership value of an element is equal to the percentage of intervals that contain the element. In contrast, as illustrated in Fig. 2a, the α -cuts invariably become progressively shorter from bottom to top. This raises questions about whether (a) shorter slices should be assigned smaller weights, or (b) the weight of an α -cut should be associated with its α value. Note that (a) and (b) represent conflicting ideas. Consequently, there seems to be no intuitively simple way to quantify the membership degree of an element directly from the collection of the α -cuts that contain it, other than perhaps appealing to non-classical notions that can be difficult to decipher.
- (ii) *Handling Sub-normal Fuzzy Numbers*: If a given fuzzy number is sub-normal (i.e., $\mu_A(x) = q$ when $b \leq x \leq c$ for some constant q less than 1, rather than being 1 as in Eq. (2.1)), then some of the α -cuts are actually empty intervals (see Fig. 2b). Consequently, performing fuzzy arithmetic (e.g., addition of fuzzy numbers)

based on or related to the α -cut approach (e.g., Boukezzoula et al., 2014; Hanss, 2005) quickly runs into difficulties: it is unclear what an appropriate outcome should be for adding a regular interval from a fuzzy number A_1 to an empty interval from a fuzzy number A_2 . In contrast, the K -intervals representation of fuzzy numbers can be adapted to handle sub-normal fuzzy numbers (not discussed here to conserve space; details described in Appendix B of Ngan (2021)), ensuring that none of the resulting K intervals is empty. This generalization allows us to handle fuzzy arithmetic of both normal and sub-normal fuzzy numbers in a unified way.

- (iii) *Validity of Membership Functions*: When applying α -cut-based approaches to two fuzzy numbers (e.g., when performing fuzzy arithmetic), α -cuts of the same α -level from the given fuzzy numbers are combined and processed to yield the α -cut of the resultant fuzzy number. However, the collection of these resultant α -cuts is not guaranteed to yield a valid membership function for the resultant fuzzy number. In fact, examples have been found showing that the resultant $\mu_{A_1 * A_2}(x)$ (where “*” is a given arithmetic operation applied to given fuzzy numbers A_1 and A_2) may produce multiple membership values for certain $x \in \mathfrak{X}$ (Boukezzoula et al., 2014; Chalco-Cano et al., 2014). That is, the resultant $\mu_{A_1 * A_2}(x)$ may actually not be a valid function, and one will need to use somewhat ad hoc heuristic schemes to turn it into a proper function. Such pathologies do not occur under the K -intervals representation, as the membership value of a given $y \in Y$ is uniquely determined as the percentage of the K intervals that contain it.

Admittedly, the K -intervals representation is just one method for decomposing a fuzzy number into slices, with the resulting slices arranged from right to left as k increases from 1 to K (as shown in Fig. 1b). While the existence of other, potentially superior decomposition schemes cannot be ruled out, we believe the K -intervals representation is among the simplest and most intuitively obvious.

2.2. An extension principle for interval analysis

The K -intervals representation introduced in the previous section enables the decomposition of a fuzzy number (a non-classical object) into a set of K intervals (a set of conceptually simpler objects). The framework discussed below allows the further decomposition of these conceptually simpler objects into very basic classic objects, leading to an extension principle for interval analysis.

Ngan (2024) uses a key idea from Chalco-Cano et al. (2014), Lodwick and Jenkins (2013) and Mizukoshi, Costa, Chalco-Cano, and Lodwick (2024) that an interval $I_A \equiv [x_A^L, x_A^R]$ can be represented as a linear function of a single variable, with a non-negative slope and defined over the domain $[0, 1]$:

Definition 2.3. (*Linear function representation of an interval*) Consider an interval $I \equiv [x_I^L, x_I^R]$. The corresponding linear function for this interval is defined as follows:

$$x_I(\lambda) = (1 - \lambda) \cdot x_I^L + \lambda \cdot x_I^R \text{ with } 0 \leq \lambda \leq 1 \quad (2.5)$$

End of Definition.

To make it more tractable, Ngan (2024) considers a discretized version of Definition 2.3:

Definition 2.4. (*The discrete set representation of an interval*) Given an interval $I \equiv [x_I^L, x_I^R]$, the discrete set representation of this interval is defined as follows:

$$\tilde{I} \equiv \{x_I^m | m = 1, 2, \dots, M\} \quad (2.6)$$

where $x_I^m = (1 - \lambda_m) \cdot x_I^L + \lambda_m \cdot x_I^R$ with $\lambda_m \equiv \frac{m-1}{M}$ and $m \in \{1, 2, \dots, M\}$ and M is a user-preselected positive integer. **End of Definition.**

To illustrate the utility of Definition 2.4, Ngan (2024) considers a situation in which a manager is contemplating a possible pay raise for an employee A: The manager deems A to be deserving of a pay raise within the range $I_A \equiv [x_{I_A}^L, x_{I_A}^R]$, but she is ultimately required to select a crisp value from I_A . It is assumed that when the manager is in a tightfisted frame of mind regarding employee A, she would recommend $x_{I_A}^L$ as his pay raise. Conversely, when she is in a generous frame of mind regarding employee A, she would recommend $x_{I_A}^R$. In summary, we can regard the crisp values $\{x_{I_A}^m | m \in \{1, \dots, M\}\}$ as representing outcomes from the M different mind-frames of a DM.

2.2.1. Matching pattern for a pair of intervals

Let us further suppose that the manager is also tasked with deciding the pay raise of another employee, B. The manager deems B worthy of a pay raise within the range $I_B \equiv [x_{I_B}^L, x_{I_B}^R]$. A key question is: if in a particular mind-frame, the manager decides upon a pay raise $x_{I_A}^L$ (or $x_{I_A}^R$) for employee A, does that imply the manager will decide upon a pay raise $x_{I_B}^L$ (or $x_{I_B}^R$) for employee B in that same mind-frame? Ngan (2024) describes a few possible scenarios:

- I. *Fair and Equal Treatment*: If the manager wishes to treat both employees fairly and equally, it could be the case that a pay raise $x_{I_A}^m$ for A is associated with a pay raise $x_{I_B}^m$ for B, i.e., a pay raise for A is proportional to a pay raise for B. (Recall that $x_{I_A}^m$ and $x_{I_B}^m$ are determined based on Eq. (2.6)).
- II. *Independent Decisions*: If the fact that the manager has selected a pay raise $x_{I_A}^m$ for A has absolutely no bearing on what she would select as the pay raise for B, we can model this situation by associating $x_{I_A}^m$ with every element in $\{x_{I_B}^{m'} | m' \in \{1, \dots, M\}\}$.
- III. *Budgetary Constraints*: If the department has certain budgetary constraints on giving pay raises to its employees, it could be the case that a pay raise $x_{I_A}^m$ for A is associated with a pay raise $x_{I_B}^{M-m+1}$ for B, i.e., a higher pay raise for A is associated with a lesser pay raise for B.

Scenarios I, II and III describe three distinct matching patterns for the values from I_A with those from I_B . In general, a matching pattern for a pair of intervals is characterized as follows:

Definition 2.5. (A matching pattern for a pair of intervals) We are given the following:

- (i) Two intervals $I_A \equiv [x_{I_A}^L, x_{I_A}^R]$ and $I_B \equiv [x_{I_B}^L, x_{I_B}^R]$, and hence their discrete set representations $\tilde{I}_A$ and $\tilde{I}_B$ (obtainable from I_A and I_B via Definition 2.4);
- (ii) A pair of values $x_{I_A}^{m_A} \in \tilde{I}_A$ and $x_{I_B}^{m_B} \in \tilde{I}_B$.

With the above, let $a_{I_A, I_B}^{m_A, m_B}$ denote the *presence* or *absence* of linkage between $x_{I_A}^{m_A}$ and $x_{I_B}^{m_B}$, by setting $a_{I_A, I_B}^{m_A, m_B}$ to 1 or 0 respectively. A set of values $\Pi_{I_A, I_B} \equiv \{a_{I_A, I_B}^{m_A, m_B} | m_A, m_B \in \{1, \dots, M\}\}$ constitutes a matching pattern for I_A and I_B if:

$$a_{I_A, I_B}^{m_A, m_B} \in \{0, 1\} \text{ for all } m_A, m_B \in \{1, \dots, M\} \quad (2.7.1)$$

$$\sum_{m_B=1}^M a_{I_A, I_B}^{m_A, m_B} \geq 1 \text{ for all } m_A \in \{1, \dots, M\} \quad (2.7.2)$$

$$\text{and } \sum_{m_A=1}^M a_{I_A, I_B}^{m_A, m_B} \geq 1 \text{ for all } m_B \in \{1, \dots, M\} \quad (2.7.3)$$

End of Definition. Thus, the matching patterns described in the aforementioned Scenarios I, II, and III can be expressed as follows:

$$1. a_{I_A, I_B}^{m_A, m_B} = \begin{cases} 1 & \text{if } m_A = m_B \\ 0 & \text{otherwise} \end{cases} \text{ for } m_A, m_B \in \{1, \dots, M\} \quad (2.8)$$

$$2. a_{I_A, I_B}^{m_A, m_B} = 1 \text{ for all } m_A, m_B \in \{1, \dots, M\} \quad (2.9)$$

$$3. a_{I_A, I_B}^{m_A, m_B} = \begin{cases} 1 & \text{if } m_B = M - m_A + 1 \\ 0 & \text{otherwise} \end{cases} \text{ for } m_A, m_B \in \{1, \dots, M\} \quad (2.10)$$

These are termed as the *perfectly correlated*, *uncorrelated*, and *perfectly anti-correlated* matching patterns for two intervals, respectively. They are illustrated in Fig. 3.

More generally, Ngan (2024) suggests the following heuristic guideline to aid users in specifying a matching pattern:

Guideline 2.1 (Specifying an appropriate matching pattern for a pair of intervals) Given a pair of intervals $I_A \equiv [x_{I_A}^L, x_{I_A}^R]$ and $I_B \equiv [x_{I_B}^L, x_{I_B}^R]$:

- (a) Obtain the corresponding discrete set representations $\tilde{I}_A \equiv \{x_{I_A}^{m_A} | m_A \in \{1, \dots, M\}\}$ and $\tilde{I}_B \equiv \{x_{I_B}^{m_B} | m_B \in \{1, \dots, M\}\}$ according to Definition 2.4. (To control complexity, choose a relatively small value for M [e.g., $M = 10$].) Initialize $\Pi_{I_A, I_B} \equiv \{a_{I_A, I_B}^{m_A, m_B} | m_A, m_B \in \{1, \dots, M\}\}$ with $a_{I_A, I_B}^{m_A, m_B} = 0$ for all $m_A, m_B \in \{1, \dots, M\}$.
- (b) For each value $x_{I_A}^{m_A}$ in $\tilde{I}_A$: Assume that this particular value $x_{I_A}^{m_A}$ has been chosen in a particular mind-frame of the stakeholder. Then ask: in the same mind-frame, what are the possible values from $\{x_{I_B}^{m_B} | m_B = 1, 2, \dots, M\}$ that the stakeholder may possibly choose? Suppose these possible values are $x_{I_B}^{m'_B}, x_{I_B}^{m''_B}$ and $x_{I_B}^{m'''_B}$. Then, update Π_{I_A, I_B} by setting $a_{I_A, I_B}^{m_A, m'_B} = a_{I_A, I_B}^{m_A, m''_B} = a_{I_A, I_B}^{m_A, m'''_B} = 1$.
- (c) Analogously, for each value $x_{I_B}^{m_B}$ in $\tilde{I}_B$: Assume that this particular value $x_{I_B}^{m_B}$ has been chosen in a particular mind-frame of the stakeholder. Then ask: in the same mind-frame, what are the possible values from $\{x_{I_A}^{m_A} | m_A = 1, 2, \dots, M\}$ that the stakeholder may possibly choose? Suppose these possible values are $x_{I_A}^{m'_A}, x_{I_A}^{m''_A}$ and $x_{I_A}^{m'''_A}$. Then, update Π_{I_A, I_B} by setting $a_{I_A, I_B}^{m'_A, m_B} = a_{I_A, I_B}^{m''_A, m_B} = a_{I_A, I_B}^{m'''_A, m_B} = 1$.

End of Guideline.

With the above guideline, a user will obtain the “first draft” of a matching pattern for I_A and I_B . If the user wishes to refine the first draft, he can choose to repeat Guideline 2.1 with progressively larger values for M .

(Supplemental note: For simplicity, we have described Guideline 2.1 in terms of the “mind-frame” of a stakeholder. Generally, to decide upon a matching pattern for $\tilde{I}_A$ and $\tilde{I}_B$, an FST user may simply ask himself: In a given setting, if he is forced to choose a particular value $x_{I_A}^{m_A}$ from $\tilde{I}_A$ [e.g., as the pay raise of employee A], what the possible value(s) would he select from $\tilde{I}_B$ [e.g., as the possible pay raise of employee B] in the same setting?).

Building upon Definition 2.5, an essential concept termed a *full specification of matching patterns for a given set of intervals* is encapsulated as follows:

Definition 2.6. (Full specification of matching patterns for a set of intervals) Let us consider a set $S = \{I_1, \dots, I_N\}$ of intervals $I_n \equiv [x_{I_n}^L, x_{I_n}^R]$, where $n = 1, \dots, N$. We view the elements of S (i.e., the intervals I_n 's) as nodes of a corresponding graph. Then:

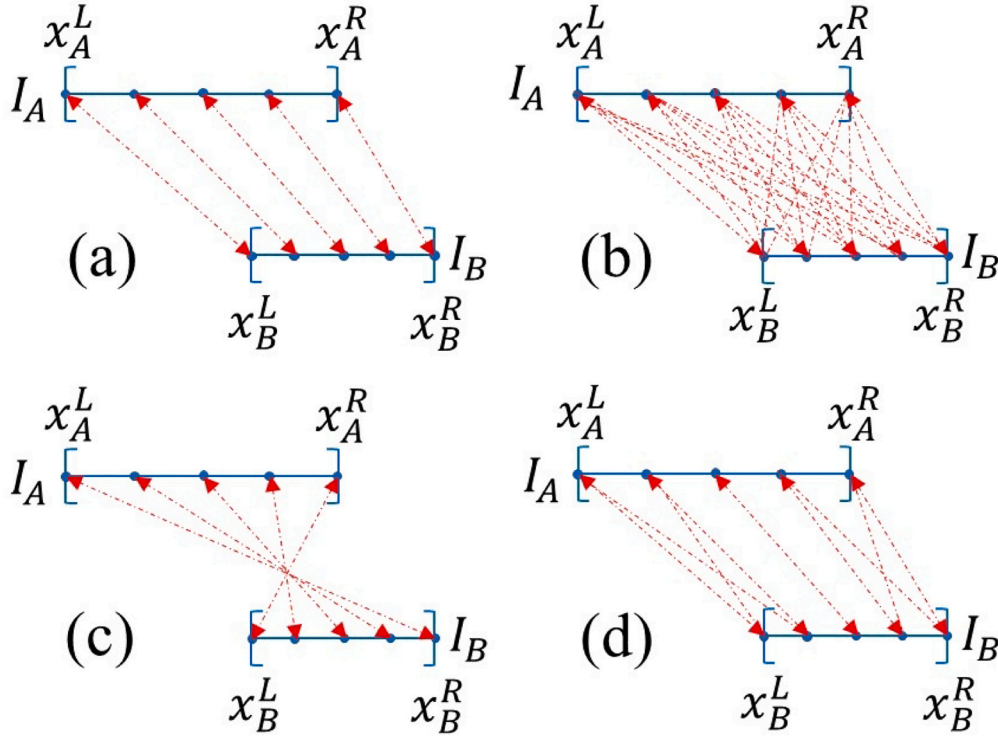


Fig. 3. Some possible matching patterns for a given pair of intervals. (a) The perfectly correlated matching pattern. (For clarity, only a few sample points from each interval are shown.) (b) The uncorrelated matching pattern. (c) The perfectly anti-correlated matching pattern. (d) A matching pattern that is a hybrid of (a) and (b).

- (i) If a matching pattern Π_{I_u, I_w} has been defined for a given pair of elements I_u and I_w from S , then I_u and I_w are considered connected with an undirected link in the graph.
- (ii) Moreover, if matching patterns have been defined for selected pairs of intervals from S such that the resultant graph is a tree (i. e., the resultant graph is connected, undirected and acyclic), then this set of matching patterns is termed a *full specification of matching patterns for S* .
- (iii) In equation form, such a full specification of matching patterns is represented as:

$$\Lambda \equiv \{\{I_1, I_2\}, \{I_2, I_3\}, \{I_3, I_4\}\} \quad (2.12)$$

Alternatively, if a matching pattern has been defined for each of the pairs of intervals $\{I_1, I_2\}$, $\{I_2, I_3\}$ and $\{I_2, I_4\}$ as depicted in Fig. 4b, the resultant graph is also a tree, corresponding to another full specification of matching patterns for S :

$$\Lambda \equiv \{\{I_1, I_2\}, \{I_2, I_3\}, \{I_2, I_4\}\} \quad (2.13)$$

On the other hand, sets of matching patterns such as

$\{\{I_1, I_2\}, \{I_3, I_4\}\}$ and $\{\{I_1, I_2\}, \{I_2, I_3\}, \{I_3, I_4\}, \{I_1, I_4\}\}$ are not full specifications for S , as the graph corresponding to the former one is not fully connected (Fig. 4c), and the graph corresponding to the latter one is not acyclic (Fig. 4d).

2.2.2. From matching patterns to an extension principle for interval analysis

Consider again the aforementioned pay-rise situation. Suppose that $f: \Re \rightarrow \Re$ is a function that maps a raise pay x to a corresponding pressure y on the department's budget. Then the set of pay raises $\tilde{I}_A \equiv \{x_{I_A}^m : m = 1, 2, \dots, M\}$ as specified in Eq. (2.6) will induce the corresponding set of budget pressures:

$$C_{I_A}^f \equiv \{f(x_{I_A}^m) | m = 1, \dots, M\} \quad (2.14)$$

Consequently, the interval J of departmental budgetary pressures resulting from I_A can be numerically approximated as

$$J \cong \left[\inf_{v \in C_{I_A}^f} \{v\}, \sup_{v \in C_{I_A}^f} \{v\} \right] \quad (2.15)$$

Encapsulating the above description, an extension principle for interval analysis for the $\Re$ case can be formulated as follows:

End of Definition. To illustrate Definition 2.6, consider a set of intervals $S = \{I_1, I_2, I_3, I_4\}$. If a matching pattern has been defined for each of the pairs of intervals $\{I_1, I_2\}$, $\{I_2, I_3\}$ and $\{I_3, I_4\}$ as depicted in Fig. 4a, then since the resultant graph is a tree, this constitutes a full specification of matching patterns for S :

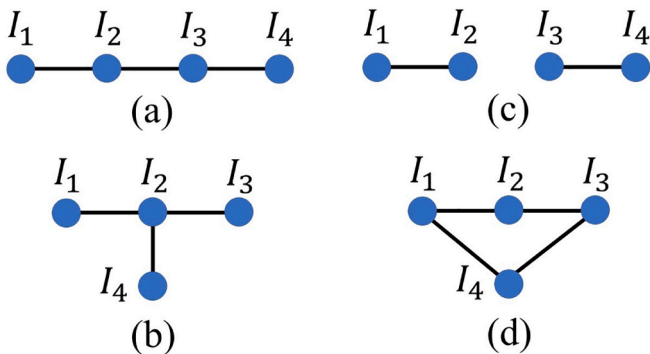


Fig. 4. Matching patterns for a set of four intervals $S = \{I_1, I_2, I_3, I_4\}$. Each of (a) and (b) is an example of full specification of matching patterns for S , as the resultant graph is a tree (i.e., connected, undirected as well as acyclic). Each of (c) and (d) is not a full specification of matching patterns for S , as (c) is not fully connected, and (d) is non-acyclic.

Definition 2.7. (An extension principle for interval analysis: the $\mathbb{N}$ case)
We are given the following:

- (i) A real-valued continuous function $f : \mathbb{N} \rightarrow \mathbb{N}$;
- (ii) An interval $I_A \equiv [x_{I_A}^L, x_{I_A}^R]$, and its discrete set representation $\tilde{I}_A \equiv \{x_{I_A}^m | m = 1, 2, \dots, M\}$ determined according to Definition 2.4, where M is a user-preselected positive integer.

With the above, we:

- (a) Form the collection of values $C_{I_A}^f$ as described in Eq. (2.14), i.e.,

$$C_{I_A}^f \equiv \{f(x_{I_A}^m) | m = 1, \dots, M\} \quad (2.16)$$

- (b) Define the interval-valued version $\bar{f}$ of f , such that upon receiving I_A as inputs, $\bar{f}(I_A)$ is numerically approximated as:

$$\bar{f}(I_A) \cong \left[\inf_{v \in C_{I_A}^f} \{v\}, \sup_{v \in C_{I_A}^f} \{v\} \right] \quad (2.17)$$

End of Definition.

For the $\mathbb{N}^2$ case (i.e., for a continuous function $f : \mathbb{N}^2 \rightarrow \mathbb{N}$), with the help of Definition 2.5, we extend Definition 2.7 to become:

Definition 2.8. (An extension principle for interval analysis: the $\mathbb{N}^2$ case)
We are given the following:

- (i) A real-valued continuous function $f : \mathbb{N}^2 \rightarrow \mathbb{N}$;
- (ii) A pair of intervals $I_A \equiv [x_{I_A}^L, x_{I_A}^R]$ and $I_B \equiv [x_{I_B}^L, x_{I_B}^R]$, and their discrete set representations $\tilde{I}_A \equiv \{x_{I_A}^m | m_A = 1, 2, \dots, M\}$ and $\tilde{I}_B \equiv \{x_{I_B}^m | m_B \in \{1, \dots, M\}\}$ determined according to Definition 2.4, where M is a user-preselected positive integer;
- (iii) A matching pattern $\Pi_{I_A, I_B} \equiv \{a_{I_A, I_B}^{m_A, m_B} | m_A, m_B \in \{1, \dots, M\}\}$ defined in accordance with Definition 2.5.

With the above, we:

- (a) Form the following collection of values:

$$C_{I_A, I_B}^f \equiv \{f(x_{I_A}^{m_A}, x_{I_B}^{m_B}) | \forall a_{I_A, I_B}^{m_A, m_B} \in \Pi_{I_A, I_B} \text{ such that } a_{I_A, I_B}^{m_A, m_B} = 1\} \quad (2.18)$$

Thus, a value $f(x_{I_A}^{m_A}, x_{I_B}^{m_B})$ belongs to C_{I_A, I_B}^f if and only if $x_{I_A}^{m_A}$ and $x_{I_B}^{m_B}$ are considered connected (i.e., if $a_{I_A, I_B}^{m_A, m_B} = 1$) according to the given matching pattern Π_{I_A, I_B} .

- (b) Define the interval-valued version $\bar{f}$ of f , such that upon receiving I_A and I_B as inputs, $\bar{f}(I_A, I_B)$ is numerically approximated as:

$$\bar{f}(I_A, I_B) \cong \left[\inf_{v \in C_{I_A, I_B}^f} \{v\}, \sup_{v \in C_{I_A, I_B}^f} \{v\} \right] \quad (2.19)$$

End of Definition.

Finally, for the $\mathbb{N}^N$ case, with the help of Definition 2.6, we extend Definition 2.8 to become:

Definition 2.9. (An extension principle for interval analysis: the $\mathbb{N}^N$ case)
We are given the following:

- (i) A real-valued continuous function $f : \mathbb{N}^N \rightarrow \mathbb{N}$, where N is a positive integer;

- (ii) A set S of intervals $I_n \equiv [x_{I_n}^L, x_{I_n}^R]$, where $n \in \{1, \dots, N\}$, and their discrete set representations $\tilde{I}_n \equiv \{x_{I_n}^m | m_n = 1, 2, \dots, M\}$ determined according to Definition 2.4, where M is a user-preselected positive integer;
- (iii) A full specification of matching patterns Λ for the set of intervals $S = \{I_1, \dots, I_N\}$ defined in accordance with Definition 2.6.

With the above, we:

- (a) Form the following collection of values:

$$C_{I_1, \dots, I_N}^f \equiv \left\{ f(x_{I_1}^{m_1}, \dots, x_{I_N}^{m_N}) \mid \forall m_1, \dots, m_N \in \{1, \dots, M\} \text{ and } \forall \{I_u, I_w\} \in \Lambda \text{ such that } a_{I_u, I_w}^{m_u, m_w} \in \Pi_{I_u, I_w} \text{ and } a_{I_u, I_w}^{m_u, m_w} = 1 \right\} \quad (2.20)$$

Thus, a value $f(x_{I_1}^{m_1}, \dots, x_{I_N}^{m_N})$ belongs to $C_{I_1, \dots, I_N}^f$ if and only if there exists an unbroken pathway linking the elements $x_{I_1}^{m_1}, \dots, x_{I_N}^{m_N}$ according to the matching patterns specified in Λ . (See an example below for an illustration.)

- (b) Define the interval-valued version $\bar{f}$ of f , such that upon receiving $I_1, \dots, I_N$ as inputs, $\bar{f}(I_1, \dots, I_N)$ is numerically approximated as:

$$\bar{f}(I_1, \dots, I_N) \cong \left[\inf_{v \in C_{I_1, \dots, I_N}^f} \{v\}, \sup_{v \in C_{I_1, \dots, I_N}^f} \{v\} \right] \quad (2.21)$$

End of Definition.

To illustrate Eq. (2.20), consider the full specification of matching patterns Λ for the set of intervals $S = \{I_1, I_2, I_3, I_4\}$ depicted in Fig. 4a. Suppose that (i) the matching pattern for each of $\{I_1, I_2\}$ and $\{I_2, I_3\}$ is of the perfectly correlated type (as described by Eq. (2.8)), i.e.,

$$\Pi_{I_u, I_w} \equiv \{a_{I_u, I_w}^{m_u, m_w} | m_u, m_w \in \{1, \dots, M\}\} \text{ with } a_{I_u, I_w}^{m_u, m_w} = \begin{cases} 1 & \text{if } m_u = m_w \\ 0 & \text{otherwise} \end{cases} \quad (2.22)$$

for $(u, w) \in \{(1, 2), (2, 3)\}$ and (ii) the matching pattern for $\{I_3, I_4\}$ is of the uncorrelated type (as described by Eq. (2.9)), i.e.,

$$\Pi_{I_3, I_4} \equiv \{a_{I_3, I_4}^{m_3, m_4} | m_3, m_4 \in \{1, \dots, M\}\} \text{ with } a_{I_3, I_4}^{m_3, m_4} = 1 \text{ for any } m_3, m_4 \in \{1, \dots, M\} \quad (2.23)$$

These give us Fig. 5a. As described in Fig. 5b, given the elements $x_{I_1}^{m_1}$, $x_{I_2}^{m_2}$, $x_{I_3}^{m_3}$ and $x_{I_4}^{m_4}$, the value $f(x_{I_1}^{m_1}, x_{I_2}^{m_2}, x_{I_3}^{m_3}, x_{I_4}^{m_4})$ is included in C_{I_1, I_2, I_3, I_4}^f if and only if there is an unbroken pathway linking $x_{I_1}^{m_1}$, $x_{I_2}^{m_2}$, $x_{I_3}^{m_3}$ and $x_{I_4}^{m_4}$ together. Such an unbroken pathway exists if and only if each of $a_{I_1, I_2}^{m_1, m_2}$, $a_{I_2, I_3}^{m_2, m_3}$ and $a_{I_3, I_4}^{m_3, m_4}$ equals one. For example, $f(x_{I_1}^4, x_{I_2}^4, x_{I_3}^4, x_{I_4}^2)$ is included in C_{I_1, I_2, I_3, I_4}^f as $a_{I_1, I_2}^{4, 4}$, $a_{I_2, I_3}^{4, 4}$ and $a_{I_3, I_4}^{4, 2}$ all equal one. In contrast, $f(x_{I_1}^2, x_{I_2}^2, x_{I_3}^1, x_{I_4}^4)$ is not included in C_{I_1, I_2, I_3, I_4}^f as although both $a_{I_1, I_2}^{2, 2}$ and $a_{I_3, I_4}^{1, 4}$ equal one, $a_{I_2, I_3}^{2, 1}$ equals zero.

Fig. 6 presents a concept map illustrating the relationships among the key concepts used in forming our interval analysis extension principle.

2.3. An extension principle for FST

For clarity, we first present a preliminary version of the FST extension principle in Section 2.3.1. We then develop additional concepts in Sections 2.3.2. Subsequently, we apply these concepts to transform the preliminary version into the finalized version of the FST extension

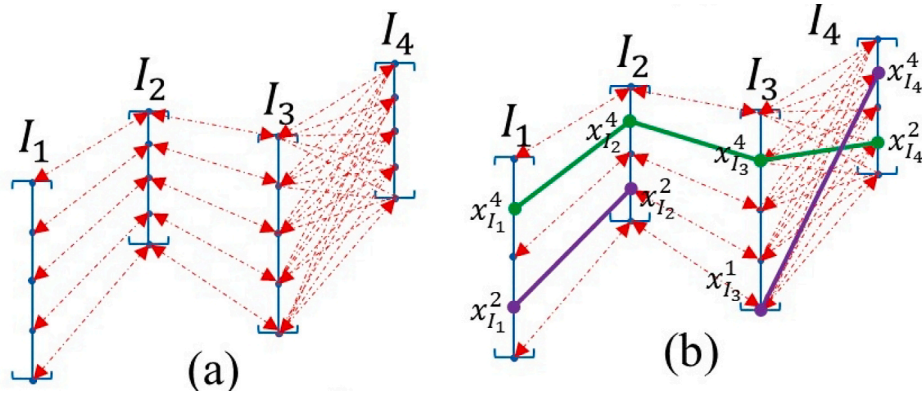


Fig. 5. (a) The matching patterns across four intervals I_1, I_2, I_3, I_4 based on Fig. 4(a) and Eqs. (2.22) and (2.23). (For clarity, only a few sample points from each interval are shown.) (b) According to (a), there is an unbroken pathway linking together the elements $x_{I_1}^{m_1}, x_{I_2}^{m_2}, x_{I_3}^{m_3}$ and $x_{I_4}^{m_4}$ when $m_1 = m_2 = m_3 = 4$ and $m_4 = 2$. Thus, the corresponding $f(x_{I_1}^4, x_{I_2}^4, x_{I_3}^4, x_{I_4}^2)$ is included in C_{I_1, I_2, I_3, I_4}^f . On the other hand, there is no unbroken pathway linking together the elements $x_{I_1}^{m_1}, x_{I_2}^{m_2}, x_{I_3}^{m_3}$ and $x_{I_4}^{m_4}$ when $m_1 = m_2 = 2, m_3 = 1$ and $m_4 = 4$. Hence, the corresponding $f(x_{I_1}^2, x_{I_2}^2, x_{I_3}^1, x_{I_4}^4)$ is not included in C_{I_1, I_2, I_3, I_4}^f .

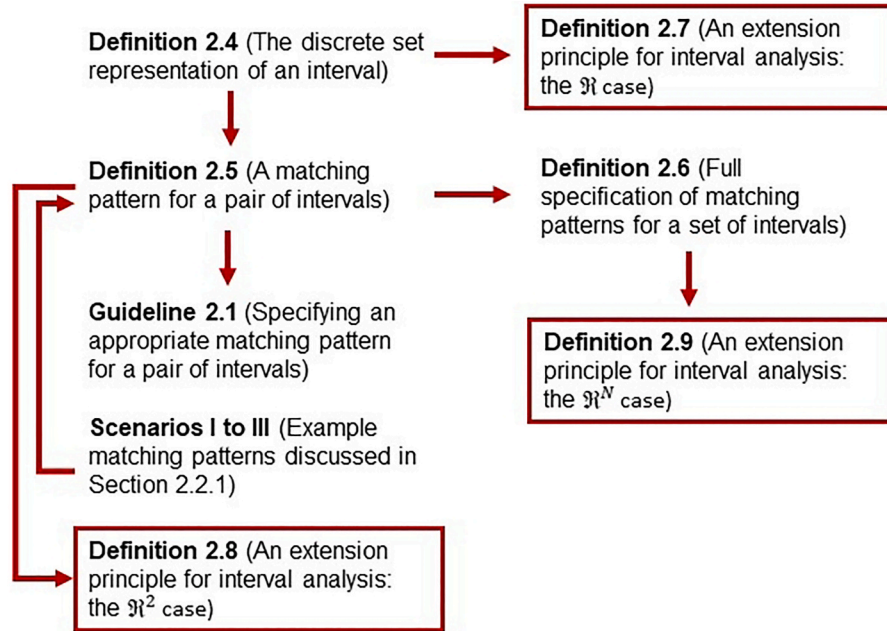


Fig. 6. A concept map illustrating the relationships among key concepts used in forming our interval analysis extension principle.

principle in Sections 2.3.3.

2.3.1. An FST extension principle: The preliminary version

Consider a continuous function $f: \mathfrak{N}^2 \rightarrow \mathfrak{N}$. For any given $(x_1, x_2) \in \mathfrak{N}^2$, the term $f(x_1, x_2)$ yields a real number. Our goal is to construct a preliminary FST extension principle that maps f to its fuzzified version $\hat{f}$: When $\hat{f}$ receives two fuzzy numbers A_1 and A_2 as its inputs, the resultant output is a fuzzy number denoted as $\hat{f}(A_1, A_2)$. This is achieved through the following steps. To cast the development into a concrete context, let us expand upon our “pay raise” example discussed in Section 2.2: Suppose that (a) A_1 and A_2 represent the “appropriate pay raises” for employees 1 and 2, respectively, that (b) there is a committee of K DMs expressing their views about the appropriate pay raises, and that (c) $\hat{f}(A_1, A_2)$ represents the resultant pressure on the department budget.

Step I: Decomposition of the given fuzzy numbers into their K -intervals representations.

Based on the K -intervals representation of fuzzy numbers described in Definition 2.2, each of A_1 and A_2 is decomposed into its associated K -

intervals representation. These decompositions produce two sets of K numerical intervals:

$$S_{A_n} \equiv \{A_n^1, A_n^2, \dots, A_n^K\} \text{ where } n \in \{1, 2\} \quad (2.24)$$

These $2 \times K$ intervals are then rearranged to form K sets of two intervals, i.e., one set of two intervals for each of the K DMs:

$$S^k \equiv \{A_1^k, A_2^k\} \text{ where } k \in \{1, \dots, K\} \quad (2.25)$$

Intuitively, $A_1^k \equiv [x_{A_1^k}^L, x_{A_1^k}^R]$ and $A_2^k \equiv [x_{A_2^k}^L, x_{A_2^k}^R]$ represent the ranges of pay raises designated by DM k as appropriate for employees 1 and 2, respectively. From Fig. 7, we observe that DM 1 is the least stringent among all DMs regarding the pay raises for employees 1 and 2. His designated intervals A_1^1 and A_2^1 are the “rightmost” among $\{A_1^k | k = 1, \dots, K\}$ and $\{A_2^k | k = 1, \dots, K\}$, respectively. The stringency of the DMs increases with k , with DM K being the most stringent among all DMs.

Step II: Using the interval analysis extension principle to process the intervals obtained from step I

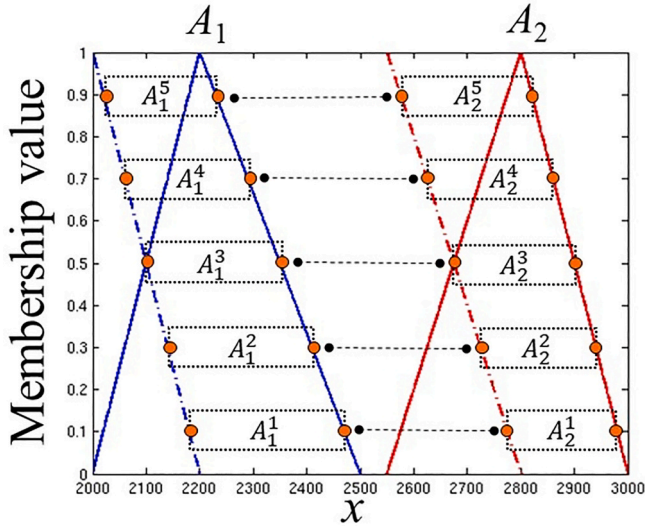


Fig. 7. Fuzzy numbers A_1 and A_2 (solid lines) representing the proposed pay raises for employees 1 and 2, respectively, along with their corresponding K -intervals representations. For illustration purpose, A_1 is decomposed into a set of 5 numerical intervals, i.e., $S_{A_1} = \{A_1^k : k = 1, \dots, 5\}$. This is similarly done for A_2 . The interval A_1^k is associated with the interval A_2^k , as both are uniquely associated with DM k .

We now apply our extension principle for interval analysis (specifically Definition 2.8) in a DM-by-DM manner, i.e., we will extend f into the interval-valued version $\tilde{f}_k$ for each $k \in \{1, \dots, K\}$: Recall that in Step I, DM k has already designated the intervals $A_1^k = [x_{A_1^k}^L, x_{A_1^k}^R]$ and $A_2^k = [x_{A_2^k}^L, x_{A_2^k}^R]$ as his chosen ranges of appropriate pay raises for employees 1 and 2.

We can readily construct the discrete set representations $\tilde{A}_1^k$ and $\tilde{A}_2^k$ of A_1^k and A_2^k based on Definition 2.4. Hence, conditions (i) and (ii) of Definition 2.8 have been fulfilled. To fulfill condition (iii) of Definition 2.8, we need to specify a matching pattern $\Pi_{A_1^k, A_2^k}$ for A_1^k and A_2^k . This task can be achieved by (1) choosing from the perfectly correlated, uncorrelated, and perfectly anti-correlated matching patterns under Scenarios I, II and III discussed in Section 2.2.1, or by (2) applying Guideline 2.1 directly to construct a matching pattern for A_1^k and A_2^k from scratch. With the matching pattern determined based on either (1) or (2), condition (iii) of Definition 2.8 is fulfilled. Hence, we can execute steps (a) and (b) of Definition 2.8. The output interval $\tilde{f}(A_1^k, A_2^k)$ is then obtained by adapting Eqs. (2.18) and (2.19) to our present situation. This $\tilde{f}(A_1^k, A_2^k)$ can be interpreted as the resultant range of budgetary effect induced by pay raises A_1^k and A_2^k according to DM k .

Step III: Using the K -intervals representation to combine the resultant intervals obtained from Step II.

We convert the set of the K intervals (i.e., $\tilde{f}(A_1^k, A_2^k)$ for $k \in \{1, \dots, K\}$) obtained in Step II into a corresponding fuzzy number by applying the key ideas contained in Eq. (2.4) of Proposition 2.1: The membership degree $\mu_{\tilde{f}(A_1, A_2)}(y)$ of a value $y \in \mathbb{R}$ with respect to $\tilde{f}(A_1, A_2)$ is to be determined as the percentage of the DMs whose intervals $\tilde{f}_k(A_1^k, A_2^k)$ contain y , i.e.,

$$\mu_{\tilde{f}(A_1, A_2)}(y) = \frac{\sum_{k=1}^K Q(y, \tilde{f}(A_1^k, A_2^k))}{K} \quad (2.26)$$

In other words, $\mu_{\tilde{f}(A_1, A_2)}(y)$ is interpreted as the percentage of the DMs who agree that y is a possible resultant budgetary effect according to their views.

We encapsulate this preliminary version of the FST extension prin-

ciple as follows:

Definition 2.10. (A preliminary version of the FST extension principle for the $\mathbb{R}^2$ case) We are given the following:

- (i) A real-valued continuous function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$;
- (ii) Two fuzzy number A_1 and A_2 , and hence their K -intervals representations $S_{A_1} \equiv \{A_1^1, A_1^2, \dots, A_1^K\}$ and $S_{A_2} \equiv \{A_2^1, A_2^2, \dots, A_2^K\}$ (obtainable from A_1 and A_2 via Definition 2.2), where K is a user-preselected positive integer.

With the above, we:

- (a) Form the following collection of intervals:

$$C_{A_1, A_2}^f \equiv \{\tilde{f}(A_1^k, A_2^k) | k \in \{1, \dots, K\}\} \quad (2.27)$$

based on Definition 2.8 (i.e., using the interval analysis extension principle for the $\mathbb{R}^2$ case);

- (b) Define the fuzzified version $\hat{f}$ of f , such that upon receiving A_1 and A_2 as inputs and for any given $y \in \mathbb{R}$, the membership function $\mu_{\hat{f}(A_1, A_2)}$ of $\hat{f}$ is computed as:

$$\mu_{\hat{f}(A_1, A_2)}(y) \equiv \sum_{I \in C_{A_1, A_2}^f} Q(y, I) / K \quad (2.28)$$

End of Definition.

2.3.2. Ingredients for developing the finalized version of an FST extension principle

We will generalize the preliminary version of the proposed FST extension principle as follows:

- (i) Instead of restricting f to be $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, we consider $f : \mathbb{R}^N \rightarrow \mathbb{R}$ with N being any positive integer. Thus, in term of our “pay raise” example, the DMs will now consider the pay raises of N employees.
- (ii) In Step I of the preliminary version, we constrained the association of the $2 \times K$ intervals with the DMs via Eq. (2.25). As a result (refer to Fig. 7), DM 1 was constrained to be the least stringent (among all DMs) for both A_1 and A_2 , and the stringency of the DMs for both A_1 and A_2 increases as k increases. However, it is conceivable that a DM who is very stringent (relative to other DMs) on one employee could be less stringent (relative to other DMs) on another employee. Thus, we will remove the type of constraints imposed by Eq. (2.25).

Similar to the preliminary version, our generalized FST extension principle consists three key steps. The main modifications occur in Step I and are made possible via the development of several essential supporting definitions. We will describe these modifications and supporting definitions in the following subsection.

2.3.2.1. Modifications to step I of the preliminary FST extension principle and development of supporting definitions. As before, we apply the K -intervals representation of fuzzy numbers (Section 2.1), this time to $A_1, A_2, \dots, A_N$, decomposing each of them into its associated K -intervals representation. These decompositions produce N set of K numerical intervals:

$$S_{A_n} \equiv \{A_n^1, A_n^2, \dots, A_n^K\} \text{ where } n \in \{1, \dots, N\} \quad (2.29)$$

Thus, S_{A_n} is the set of K intervals derived from the fuzzy number A_n .

For definiteness, we use the intervals $\{A_1^1, \dots, A_1^K\}$ obtained from A_1 to

index our DMs: A_1^k is to be associated with DM k . Due to this indexing, the least stringent of all DMs with respect to employee 1 is labelled as DM 1, whereas the most stringent of all DMs with respect to employee 1 is labelled as DM K .

A key question at this point is: How do we associate $\{A_2^k | k \in \{1, \dots, K\}\}$ obtained from A_2 with the DMs? Some possible schemes are as follows:

- We assume that DM k designates A_2^k as his view about the range of pay raise suitable for employee 2. Equivalently speaking, under this assumption, A_2^k is to be associated with A_1^k for $k \in \{1, \dots, K\}$ (as both A_1^k and A_2^k are associated with DM k under this assumption).
- We assume the fact that DM k chooses A_1^k for employee 1 tells us nothing about what he chooses for employee 2, i.e., he could have possibly chosen any of $\{A_2^k | k \in \{1, \dots, K\}\}$. Thus, under this assumption, A_1^k is to be associated with every element in $\{A_2^k | k \in \{1, \dots, K\}\}$.
- We assume the fact that DM k chooses A_1^k for employee 1 implies that he chooses A_2^{K-k+1} for employee 2. Thus, A_2^{K-k+1} is to be associated with A_1^k under this assumption.

Notice that Schemes (a), (b) and (c) are analogous to Scenarios I, II and III described in Section 2.2.1. The current schemes deal with matching patterns *at the level of the DMs* – specifically, across two sets of intervals $S_{A_1} = \{A_1^k | k \in \{1, \dots, K\}\}$ and $S_{A_2} = \{A_2^k | k \in \{1, \dots, K\}\}$ arising from a pair of given fuzzy numbers A_1 and A_2 . In contrast, the former scenarios concern matching patterns *at the level of mind-frames* – specifically, across two sets of points $\tilde{I}_A = \{x_A^m | m = 1, 2, \dots, M\}$ and $\tilde{I}_B = \{x_B^m | m = 1, 2, \dots, M\}$ arising from a pair of given intervals I_A and I_B .

The above analogy naturally leads to the adaptation of Definitions 2.5 and 2.6, originally designed for dealing with matching patterns at the mind-frame level, to form corresponding definitions for dealing with matching patterns at the DM level. Specifically, the DM-level version of Definitions 2.5 is:

Definition 2.5' (A matching pattern for a pair of fuzzy numbers) We are given the following:

- Two fuzzy numbers A_1 and A_2 , and hence their K -intervals representations $S_{A_1} \equiv \{A_1^1, A_1^2, \dots, A_1^K\}$ and $S_{A_2} \equiv \{A_2^1, A_2^2, \dots, A_2^K\}$ (obtainable from A_1 and A_2 via Definition 2.2), where K is a user-preselected positive integer;
- A pair of intervals $A_1^{k_1} \in S_1$ and $A_2^{k_2} \in S_2$.

With the above, let $a_{A_1, A_2}^{k_1, k_2}$ denote the *presence* or *absence* of linkage between $A_1^{k_1}$ and $A_2^{k_2}$, by setting $a_{A_1, A_2}^{k_1, k_2}$ to 1 or 0 respectively. A set of values $\Pi_{A_1, A_2} \equiv \{a_{A_1, A_2}^{k_1, k_2} | k_1, k_2 \in \{1, \dots, K\}\}$ constitutes a matching pattern for A_1 and A_2 if

$$a_{A_1, A_2}^{k_1, k_2} \in \{0, 1\} \text{ for all } k_1, k_2 \in \{1, \dots, K\} \quad (2.30.1)$$

$$\sum_{k_2=1}^K a_{A_1, A_2}^{k_1, k_2} \geq 1 \text{ for all } k_1 \in \{1, \dots, K\} \quad (2.30.2)$$

$$\text{and } \sum_{k_1=1}^K a_{A_1, A_2}^{k_1, k_2} \geq 1 \text{ for all } k_2 \in \{1, \dots, K\} \quad (2.30.3)$$

End of Definition.

Thus, the matching patterns described in Schemes (a), (b) and (c) can be expressed as:

$$1. a_{A_1, A_2}^{k_1, k_2} = \begin{cases} 1 & \text{if } k_1 = k_2 \\ 0 & \text{otherwise} \end{cases} \text{ for } k_1, k_2 \in \{1, \dots, K\} \quad (2.31)$$

$$2. a_{A_1, A_2}^{k_1, k_2} = 1 \text{ for all } k_1, k_2 \in \{1, \dots, K\} \quad (2.32)$$

$$3. a_{A_1, A_2}^{k_1, k_2} = \begin{cases} 1 & \text{if } k_2 = K - k_1 + 1 \\ 0 & \text{otherwise} \end{cases} \text{ for } k_1, k_2 \in \{1, \dots, K\} \quad (2.33)$$

These are denoted as the *perfectly correlated*, *uncorrelated*, and *perfectly anti-correlated* matching patterns for a pair of fuzzy numbers, respectively. They are illustrated in Fig. 8.

Furthermore, to specify a matching pattern across a pair of fuzzy numbers *from scratch*, we adapt Guideline 2.1, originally developed for handling matching patterns at the mind-frame level, to become the corresponding guideline for handling matching patterns at the DM level:

Guideline 2.1' (Specifying an appropriate matching pattern for a pair of fuzzy numbers) Given a pair of fuzzy numbers A_1 and A_2 :

- Obtain the corresponding K -intervals decompositions $S_{A_1} \equiv \{A_1^1, A_1^2, \dots, A_1^K\}$ and $S_{A_2} \equiv \{A_2^1, A_2^2, \dots, A_2^K\}$ according to Definition 2.2. Initialize $\Pi_{A_1, A_2} \equiv \{a_{A_1, A_2}^{k_1, k_2} | k_1, k_2 \in \{1, \dots, K\}\}$ with $a_{A_1, A_2}^{k_1, k_2} = 0$ for all $k_1, k_2 \in \{1, \dots, K\}$.
- For each interval $A_1^{k_1}$ in S_{A_1} : Assume that this particular interval $A_1^{k_1}$ has been designated by a certain DM as his view regarding A_1 . Then ask: for the same DM, what are the possible intervals from $S_{A_2} = \{A_2^1, A_2^2, \dots, A_2^K\}$ that the DM may possibly designate as his view regarding A_2 ? Suppose these possible intervals are $A_2^{k_2'}, A_2^{k_2''}$ and $A_2^{k_2''}$. Then, update Π_{A_1, A_2} by setting $a_{A_1, A_2}^{k_1, k_2'} = a_{A_1, A_2}^{k_1, k_2''} = a_{A_1, A_2}^{k_1, k_2''} = 1$.
- Analogously, for each interval $A_2^{k_2}$ in S_{A_2} : Assume that this particular interval $A_2^{k_2}$ has been designated by a certain DM as his view regarding A_2 . Then ask: for the same DM, what are the possible intervals from $S_{A_1} = \{A_1^1, A_1^2, \dots, A_1^K\}$ that the DM may possibly designate as his view regarding A_1 ? Suppose these possible intervals are $A_1^{k_1'}, A_1^{k_1''}$ and $A_1^{k_1''}$. Then, update Π_{A_1, A_2} by setting $a_{A_1, A_2}^{k_1', k_2} = a_{A_1, A_2}^{k_1'', k_2} = a_{A_1, A_2}^{k_1'', k_2} = 1$.

End of Guideline.

(Supplemental note: For simplicity, we have described Guideline 2.1' in terms of the "view" of a DM. Generally, to decide upon a matching pattern for A_1 and A_2 , an FST user may simply ask himself: In a given setting, if he is forced to choose a particular interval $A_1^{k_1}$ from S_{A_1} to represent his view about the concept A_1 , what possible interval(s) would he select from S_{A_2} to represent his view about the concept A_2 in the same setting?).

Meanwhile, the DM-level version of Definitions 2.6 is:

Definition 2.6' (Full specification of matching patterns for a set of fuzzy numbers) We are given a set of fuzzy numbers $\Omega \equiv \{A_1, \dots, A_N\}$. We view the elements of Ω (i.e., the fuzzy numbers A_n 's) as nodes of a corresponding graph. Then:

- If a matching pattern Π_{A_u, A_w} has been defined for a given pair of elements A_u and A_w from Ω , then A_u and A_w are considered connected with an undirected link in the graph.
- Moreover, if matching patterns have been defined for selected pairs of fuzzy numbers from Ω such that the resultant graph is a tree, then this set of matching patterns is termed a *full specification of matching patterns* for Ω .
- In equation form, such a full specification of matching patterns is represented as:

$$Y \equiv \{A_u, A_w\} | \forall A_u, A_w \in \Omega \text{ such that } \Pi_{A_u, A_w} \text{ has been defined} \} \quad (2.34)$$

End of Definition.

The illustration of Definition 2.6' is similar to that of Definition 2.6: Consider a set of fuzzy numbers $\Omega \equiv \{A_1, \dots, A_4\}$. Each of the following

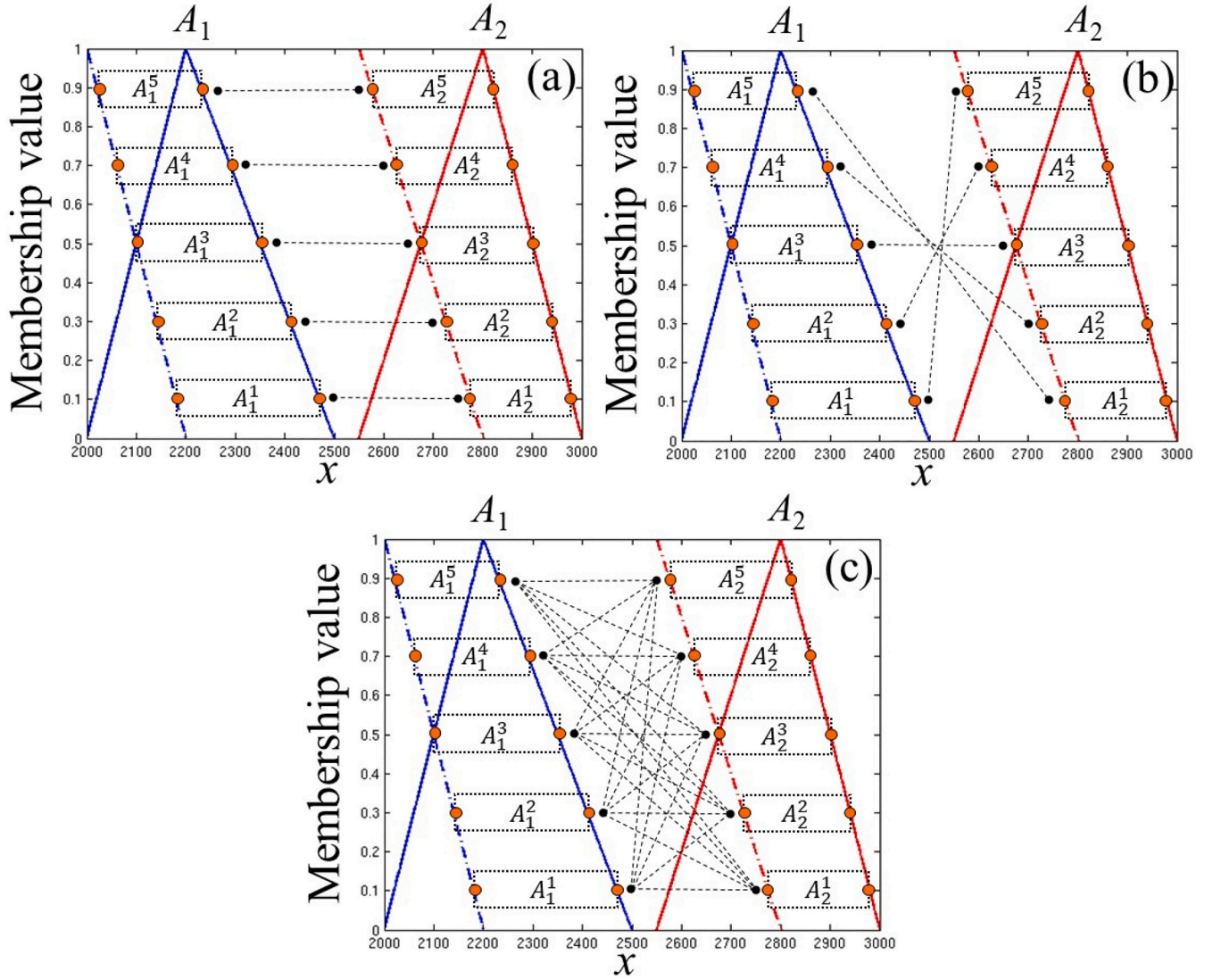


Fig. 8. Some possible matching patterns for a given pair of fuzzy numbers. (a) The perfectly correlated matching pattern. (For clarity, each fuzzy number is decomposed into a set of 5 numerical intervals only.) (b) The perfectly anti-correlated matching pattern. (c) The uncorrelated matching pattern.

two examples constitutes a full specification of matching patterns for Ω as the resultant graph is a tree:

$$Y \equiv \{\{A_1, A_2\}, \{A_2, A_3\}, \{A_3, A_4\}\} \text{ and } Y \equiv \{\{A_1, A_2\}, \{A_2, A_3\}, \{A_2, A_4\}\}$$

while the following two are not:

$$\{\{A_1, A_2\}, \{A_3, A_4\}\} \text{ and } \{\{A_1, A_2\}, \{A_2, A_3\}, \{A_3, A_4\}, \{A_1, A_4\}\}$$

as the former one is not fully connected, and the latter one is not acyclic.

2.3.3. An FST extension principle: The finalized version

Using the concepts developed in Sections 2.3.1 and 2.3.2, we are now ready to construct the finalized version of the proposed FST extension principle. This finalized version is encapsulated in the following three definitions to highlight its analogy with the extension principle for interval analysis, as described in Definitions 2.7, 2.8 and 2.9:

Definition 2.11. (An FST extension principle: the $\mathfrak{N}$ case) We are given the following:

- (i) A real-valued continuous function $f: \mathfrak{N} \rightarrow \mathfrak{N}$;

- (ii) A fuzzy number A and its K -intervals representation $S_A \equiv \{A^1, A^2, \dots, A^K\}$, determined according to Definition 2.2, where K is a user-preselected positive integer.

With the above, we:

- (a) Form the collection of intervals

$$\tilde{C}_A^f \equiv \{\tilde{f}(A^k) | k = 1, \dots, K\} \quad (2.35)$$

based on Definition 2.7 (i.e., the interval analysis extension principle for the $\mathfrak{N}$ case). Specifically, as f and the A^k 's are given, we can use Definition 2.7 to determine $\tilde{f}(A^k)$ for each $k \in \{1, \dots, K\}$.

- (b) Rename the elements of $\tilde{C}_A^f$ by defining C_A^f such that

$$C_A^f \equiv \{\tilde{f}_k | k \in \{1, \dots, K\}\} \text{ where } \tilde{f}_k \equiv \tilde{f}(A^k) \quad (2.36)$$

to emphasize that the interval $\tilde{f}_k$ is associated with DM k ;

- (c) Define the fuzzified version $\hat{f}$ of f , such that upon receiving A as input, the membership function $\mu_{\hat{f}(A)}$ of $\hat{f}$ is determined as:

$$\mu_{\tilde{f}(A)}(y) \equiv \sum_{I \in \mathcal{I}_A} Q(y, I) / K \text{ where } y \in \mathfrak{N} \quad (2.37)$$

End of Definition.

For the $\mathfrak{N}^2$ case, with the help of Definition 2.5' (i.e., the DM-level version of Definition 2.5, described in Section 2.3.2.1), we extend Definition 2.11 to become:

Definition 2.12. (An FST extension principle: the $\mathfrak{N}^2$ case) We are given the following:

- (i) A real-valued continuous function $f : \mathfrak{N}^2 \rightarrow \mathfrak{N}$;
- (ii) A pair of fuzzy number A_1 and A_2 and their K -intervals representations $S_{A_1} \equiv \{A_1^1, A_1^2, \dots, A_1^K\}$ and $S_{A_2} \equiv \{A_2^1, A_2^2, \dots, A_2^K\}$ (obtained via Definition 2.2), where K is a user-preselected positive integer;
- (iii) A matching pattern $\Pi_{A_1, A_2} \equiv \{a_{A_1, A_2}^{k_1, k_2} | k_1, k_2 \in \{1, \dots, K\}\}$ defined in accordance with Definition 2.5' in Section 2.3.2.1.

With the above, we

- (a) Form the following collection of intervals:

$$\tilde{C}_{A_1, A_2}^f \equiv \{\tilde{f}(A_1^{k_1}, A_2^{k_2}) | \forall a_{A_1, A_2}^{k_1, k_2} \in \Pi_{A_1, A_2} \text{ such that } a_{A_1, A_2}^{k_1, k_2} = 1\} \quad (2.38)$$

based on Definition 2.8 (i.e., the interval analysis extension principle for the $\mathfrak{N}^2$ case). Specifically, as f , the A_1^k 's and the A_2^k 's are given, we can use Definition 2.8 repeatedly to determine each of the $\tilde{f}(A_1^{k_1}, A_2^{k_2})$'s required in Eq. (2.38). Thus, an interval $\tilde{f}(A_1^{k_1}, A_2^{k_2})$ belongs to $\tilde{C}_{A_1, A_2}^f$ if and only if $A_1^{k_1}$ and $A_2^{k_2}$ are considered connected (i.e., if $a_{A_1, A_2}^{k_1, k_2} = 1$) according to the given matching pattern Π_{A_1, A_2} .

- (b) Rearrange and combine the elements of $\tilde{C}_{A_1, A_2}^f$ by defining $C_{A_1, A_2}^{f,1}$ and $C_{A_1, A_2}^{f,2}$ such that

$$C_{A_1, A_2}^{f,1} \equiv \{\tilde{f}_{k_1} | k_1 \in \{1, \dots, K\}\} \quad (2.39)$$

$$\text{where } \tilde{f}_{k_1} \equiv \bigcup \left\{ \tilde{f}(A_1^{k_1}, A_2^{k_2}) \mid \forall k_2 \in \{1, \dots, K\} \text{ such that } \tilde{f}(A_1^{k_1}, A_2^{k_2}) \in \tilde{C}_{A_1, A_2}^f \right\}$$

$$\text{and } C_{A_1, A_2}^{f,2} \equiv \{\tilde{f}_{k_2} | k_2 \in \{1, \dots, K\}\} \quad (2.40)$$

where

$$\tilde{f}_{k_2} \equiv \bigcup \left\{ \tilde{f}(A_1^{k_1}, A_2^{k_2}) \mid \forall k_1 \in \{1, \dots, K\} \text{ such that } \tilde{f}(A_1^{k_1}, A_2^{k_2}) \in \tilde{C}_{A_1, A_2}^f \right\}$$

- (c) Define the fuzzified version $\hat{f}$ of f , such that upon receiving A_1 and A_2 as inputs, the membership function $\mu_{\hat{f}(A_1, A_2)}$ of $\hat{f}$ is computed as:

$$\mu_{\hat{f}(A_1, A_2)}(y) \equiv \frac{1}{2} \left(\sum_{I \in C_{A_1}^{f,1}} Q(y, I) / K + \sum_{I \in C_{A_2}^{f,2}} Q(y, I) / K \right) \text{ where } y \in \mathfrak{N} \quad (2.41)$$

End of Definition.

(Supplemental note: Recall that in Section 2.3.2.1, we used the intervals $\{A_1^1, \dots, A_1^K\}$ obtained from A_1 to index our DMs, with A_1^k associated with DM k . This A_1 -centric method of DM-indexing is captured by Eq. (2.39). In the above definition, we specifically constructed Eq. (2.40) to capture the corresponding A_2 -centric method of DM-indexing.

Together, Eqs. (2.39) and (2.40) allow us to define $\mu_{\tilde{f}(A_1, A_2)}$ in Eq. (2.41) as the average of the two methods, thereby avoiding any asymmetric preference for either the A_1 - or the A_2 -centric method.).

As an illustration of the above definition, under Scheme (a) (i.e., the perfectly correlated matching pattern described in Eq. (2.31)), Eqs. (2.38), (2.39) and (2.40) would respectively become:

$$\tilde{C}_{A_1, A_2}^f = \{\tilde{f}(A_1^k, A_2^k) | k \in \{1, \dots, K\}\}$$

$$\text{and } C_{A_1, A_2}^{f,1} = C_{A_1, A_2}^{f,2} = \{\tilde{f}_k | k \in \{1, \dots, K\}\} \text{ with } \tilde{f}_k = \tilde{f}(A_1^k, A_2^k).$$

Under Scheme (b) (i.e., the uncorrelated matching pattern described in Eq. (2.32)), Eqs. (2.38), (2.39) and (2.40) would instead become:

$$\tilde{C}_{A_1, A_2}^f = \{\tilde{f}(A_1^{k_1}, A_2^{k_2}) | k_1, k_2 \in \{1, \dots, K\}\}$$

$$C_{A_1, A_2}^{f,1} = \{\tilde{f}_{k_1} | k_1 \in \{1, \dots, K\}\} \text{ with } \tilde{f}_{k_1} = \bigcup \{\tilde{f}(A_1^{k_1}, A_2^{k_2}) | k_2 \in \{1, \dots, K\}\}$$

and

$$C_{A_1, A_2}^{f,2} = \{\tilde{f}_{k_2} | k_2 \in \{1, \dots, K\}\} \text{ with } \tilde{f}_{k_2} = \bigcup \{\tilde{f}(A_1^{k_1}, A_2^{k_2}) | k_1 \in \{1, \dots, K\}\}.$$

Under Scheme (c) (i.e., the perfectly anti-correlated matching pattern described in Eq. (2.33)), Eqs. (2.38), (2.39) and (2.40) would become:

$$\tilde{C}_{A_1, A_2}^f = \{\tilde{f}(A_1^k, A_2^{K-k+1}) | k \in \{1, \dots, K\}\}$$

$$\text{and } C_{A_1, A_2}^{f,1} = C_{A_1, A_2}^{f,2} = \{\tilde{f}_k | k \in \{1, \dots, K\}\} \text{ with } \tilde{f}_k = \tilde{f}(A_1^k, A_2^{K-k+1}).$$

Finally, for the $\mathfrak{N}^N$ case, with the help of Definitions 2.5' and 2.6' (i.e., the DM-level versions of Definitions 2.5 and 2.6, described in Section 2.3.2.1), we extend Definition 2.12 to become:

Definition 2.13. (An FST extension principle: the $\mathfrak{N}^N$ case) We are given the following:

- (i) A real-valued continuous function $f : \mathfrak{N}^N \rightarrow \mathfrak{N}$, where N is a positive integer;
- (ii) A set of N fuzzy numbers $\Omega \equiv \{A_1, \dots, A_N\}$ and their K -intervals representations $S_{A_1} \equiv \{A_1^1, A_1^2, \dots, A_1^K\}, \dots, S_{A_N} \equiv \{A_N^1, A_N^2, \dots, A_N^K\}$ (obtained via Definition 2.2), where K is a user-preselected positive integer;
- (iii) A full specification of matching patterns Y for the set of fuzzy numbers $\Omega \equiv \{A_1, \dots, A_N\}$ defined in accordance with Definitions 2.6' in Section 2.3.2.1.

With the above, we:

- (a) Form the following collection of intervals:

$$\begin{aligned} \tilde{C}_{A_1, \dots, A_N}^f &\equiv \{\tilde{f}(A_1^{k_1}, \dots, A_N^{k_N}) | \forall k_1, \dots, k_N \in \{1, \dots, K\} \text{ and } \forall \{A_u, A_w\} \\ &\in Y \text{ such that } a_{A_u, A_w}^{k_u, k_w} \in \Pi_{A_u, A_w} \text{ and } a_{A_u, A_w}^{k_u, k_w} = 1\} \end{aligned} \quad (2.42)$$

based on Definition 2.9 (i.e., the interval analysis extension principle for the $\mathfrak{N}^N$ case). Specifically, as f , the A_1^k 's, ..., the A_N^k 's are given, we can use Definition 2.9 repeatedly to determine each of the $\tilde{f}(A_1^{k_1}, \dots, A_N^{k_N})$'s required in Eq. (2.42). Thus, an interval $\tilde{f}(A_1^{k_1}, \dots, A_N^{k_N})$ belongs to $\tilde{C}_{A_1, \dots, A_N}^f$ if and only if there exists an unbroken pathway linking the elements $A_1^{k_1}, \dots, A_N^{k_N}$ according to the matching patterns specified in Y . (See an example below for an illustration.).

- (b) Rearrange and combine the elements of $\tilde{C}_{A_1, \dots, A_N}^f$ by defining

$$C_{A_1, \dots, A_N}^{f,1}, C_{A_1, \dots, A_N}^{f,2}, \dots, C_{A_1, \dots, A_N}^{f,N} \text{ such that}$$

$$C_{A_1 \dots A_N}^{f,1} \equiv \left\{ \bar{f}_{k_1} | k_1 \in \{1, \dots, K\} \right\} \quad (2.43)$$

$$\text{with } \bar{f}_{k_1} \equiv \cup \left\{ \bar{f} \left(A_1^{k_1}, A_2^{k_2}, \dots, A_N^{k_N} \right) | \forall k_2, \dots, k_N \right. \\ \left. \in \{1, \dots, K\} \text{ such that } \bar{f} \left(A_1^{k_1}, A_2^{k_2}, \dots, A_N^{k_N} \right) \in \tilde{C}_{A_1 \dots A_N}^f \right\}$$

$$\text{and } C_{A_1 \dots A_N}^{f,2} \equiv \left\{ \bar{f}_{k_2} | k_2 \in \{1, \dots, K\} \right\} \quad (2.44)$$

$$\text{with } \bar{f}_{k_2} \equiv \cup \left\{ \bar{f} \left(A_1^{k_1}, A_2^{k_2}, \dots, A_N^{k_N} \right) | \forall k_1, k_3, k_4, \dots, k_N \right. \\ \left. \in \{1, \dots, K\} \text{ such that } \bar{f} \left(A_1^{k_1}, A_2^{k_2}, \dots, A_N^{k_N} \right) \in \tilde{C}_{A_1 \dots A_N}^f \right\}$$

and so on.

- (c) Define the fuzzified version $\hat{f}$ of f , such that upon receiving $A_1, \dots, A_N$ as inputs and for any $y \in \mathfrak{N}$, the membership function $\mu_{\hat{f}(A_1, \dots, A_N)}$ of $\hat{f}$ is computed as:

$$\mu_{\hat{f}(A_1, \dots, A_N)}(y) \equiv \left(\sum_{I \in C_{A_1 \dots A_N}^{f,1}} Q(y, I) / K + \sum_{I \in C_{A_1 \dots A_N}^{f,2}} Q(y, I) / K + \dots \right. \\ \left. + \sum_{I \in C_{A_1 \dots A_N}^{f,N}} Q(y, I) / K \right) / N \quad (2.45)$$

End of Definition. Note that Eqs. (2.42)–(2.45) in Definition 2.13 are the generalized versions of Eqs. (2.38)–(2.41) in Definition 2.12, as we transition from the $\mathfrak{N}^2$ case to the $\mathfrak{N}^N$ case.

To illustrate Eq. (2.42), suppose that $\Omega = \{A_1, A_2, A_3, A_4\}$, and the full specification of matching patterns for Ω is

$$Y \equiv \{\{A_1, A_2\}, \{A_2, A_3\}, \{A_3, A_4\}\} \quad (2.46)$$

Let us further suppose that (i) the matching pattern for each of $\{A_1, A_2\}$ and $\{A_3, A_4\}$ is of the uncorrelated type (as described by Eq. (2.32)), i.e.,

$$\Pi_{A_u A_w} \equiv \left\{ a_{A_u A_w}^{k_u, k_w} | k_u, k_w \in \{1, \dots, K\} \right\} \text{ with } a_{A_u A_w}^{k_u, k_w} = 1 \quad (2.47)$$

for any $k_u, k_w \in \{1, \dots, K\}$ and for $(u, w) \in \{(1, 2), (3, 4)\}$, and (ii) the

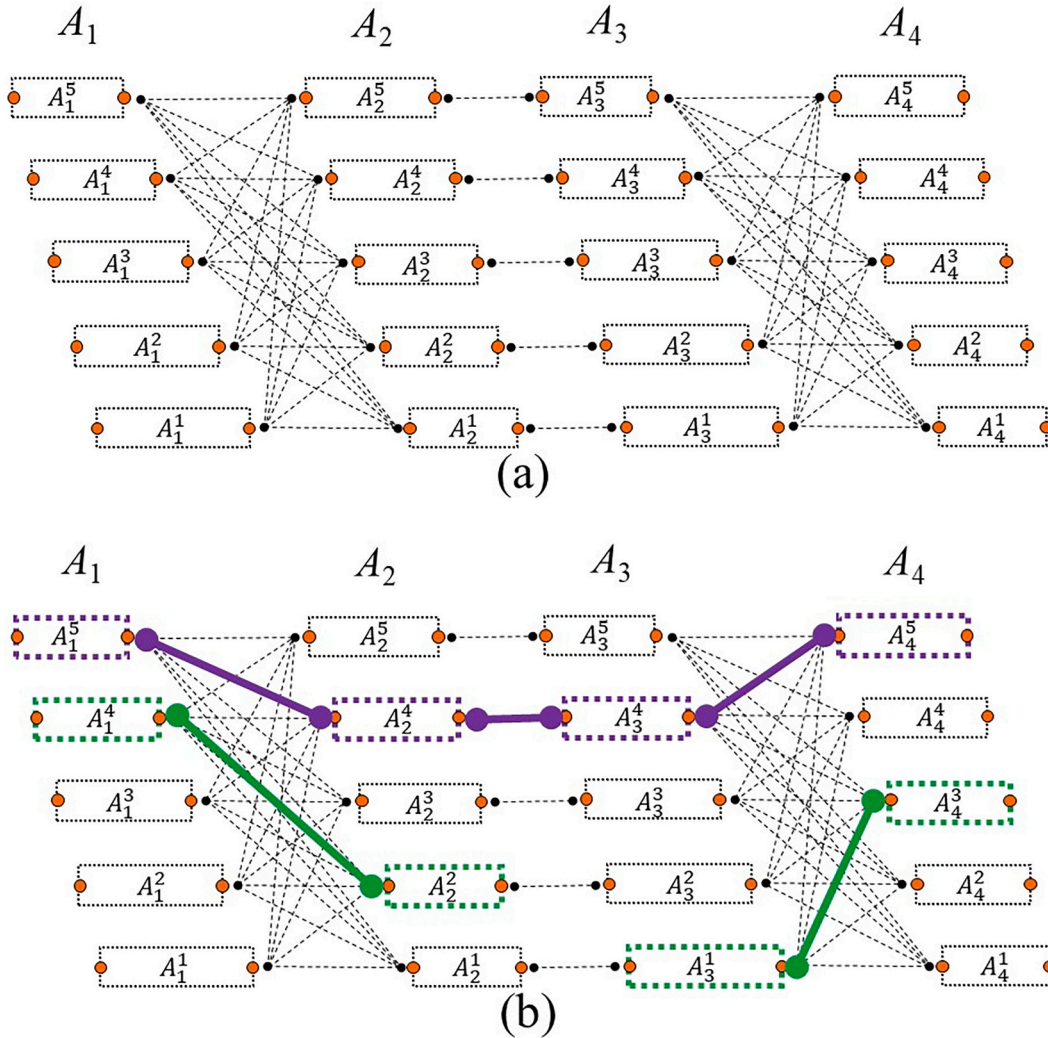


Fig. 9. (a) The matching patterns across four fuzzy numbers A_1, A_2, A_3, A_4 based on Eqs. (2.46)–(2.48). (For clarity, each fuzzy number is decomposed into a set of 5 numerical intervals only.) (b) According to (a), there is an unbroken pathway linking together the elements A_1^5, A_2^4, A_3^3 and A_4^5 . Thus, the corresponding $\bar{f}(A_1^5, A_2^4, A_3^3, A_4^5)$ is included in $\tilde{C}_{A_1 \dots A_4}^f$. On the other hand, there is no unbroken pathway linking together the elements A_1^4, A_2^3, A_3^2 and A_4^1 . Hence, the corresponding $\bar{f}(A_1^4, A_2^3, A_3^2, A_4^1)$ is not included in $\tilde{C}_{A_1 \dots A_4}^f$.

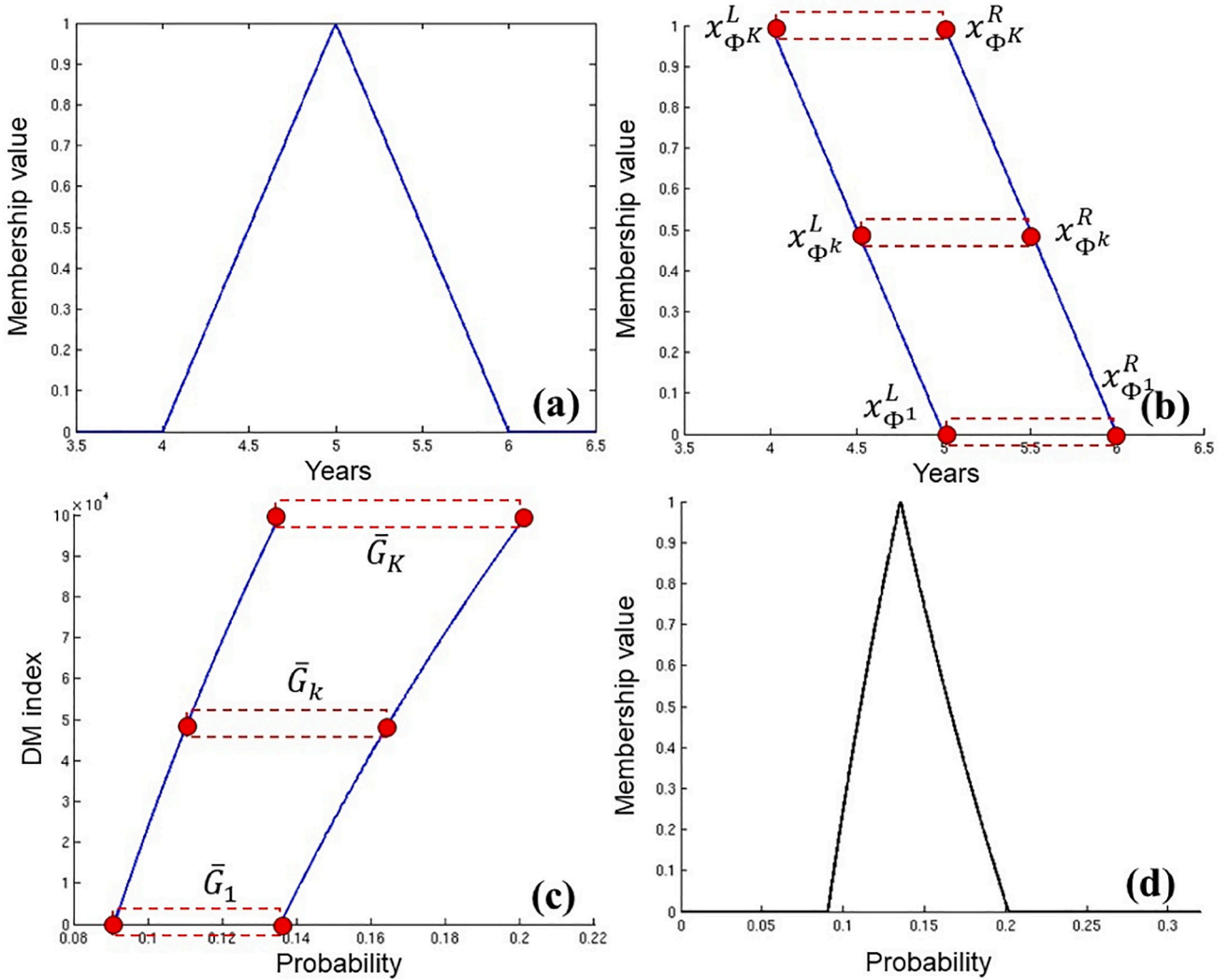


Fig. 11. Calculation of the lifetime of a piece of equipment. (a) The membership function of the fuzzy number “about 5 years”. (b) The K -intervals representation of “about 5 years”. (c) The resultant probability intervals, with interval $\bar{G}_k$ associated with DM k . (d) The membership function $\mu_{\bar{G}(\Phi)}$ of the resultant fuzzy-valued probability $\bar{G}(\Phi)$ that the piece of equipment will have a lifetime longer than “about 5 years”.

refer to Fig. 12 to view the main flow of our solution process for this problem, and to refer to Definition 2.11 in Section 2.3.3 for additional details.

Note that the key term f in Definition 2.11 correspond to the function G in this problem. Thus, condition (i) of Definition 2.11 has been fulfilled. As indicated in condition (ii) of Definition 2.11, the next step for solving this problem is to convert Φ (corresponding to the fuzzy number A in Definition 2.11) into its K -intervals representation S_Φ . As a result, the k -th numerical interval (i.e., the interval chosen by DM k describing his view of “about 5 years”) is:

$$\Phi^k = [x_{\Phi^k}^L, x_{\Phi^k}^R] = \left[5 - \left(k - \frac{1}{2}\right) / K, 6 - \left(k - \frac{1}{2}\right) / K \right] \quad (3.3)$$

as displayed in Fig. 11b. This means that according to DM k , his view of “about 5 years” can take on any point $x \in \Phi^k$.

Subsequently, we execute parts (a) and (b) of Definition 2.11 to form:

$$C_\Phi^G \equiv \{\bar{G}_k | k = 1, \dots, K\} \text{ with } \bar{G}_k = \bar{G}(\Phi^k)$$

by applying the interval analysis extension principle for the $\Re$ case (i.e., Definition 2.7) to each DM k , for $k \in \{1, \dots, K\}$. The application of Definition 2.7 leads us to the following: The probability that the equipment will have a time-to-failure of longer than “about 5 years”, according to

DM k , corresponds to the interval $\bar{G}(\Phi^k)$ numerically approximated by

$$\bar{G}(\Phi^k) \cong \left[\inf_{v \in C_{\Phi^k}^G} \{v\}, \sup_{v \in C_{\Phi^k}^G} \{v\} \right] \quad (3.4.1)$$

$$\text{with } C_{\Phi^k}^G \equiv \{G(x_{\Phi^k}^m) | m = 1, \dots, M\} \quad (3.4.2)$$

$$\text{and } x_{\Phi^k}^m = (1 - \lambda) \cdot x_{\Phi^k}^L + \lambda \cdot x_{\Phi^k}^R \text{ with } \lambda = \frac{m - \frac{1}{2}}{M} \quad (3.4.3)$$

Note that the terms $\bar{G}(\Phi^k)$ in Eq. (3.4.1) and $C_{\Phi^k}^G$ in Eq. (3.4.2) correspond to the terms $\bar{f}(I_A)$ in Eq. (2.17) and $C_{I_A}^G$ in Eqs. (2.16) of Definition 2.7, respectively. In fact, for the present problem, Eq. (3.4) can be calculated analytically by combining with Eqs. (3.1) and (3.3) and letting $M \rightarrow \infty$, resulting in:

$$\begin{aligned} \bar{G}_k &= \bar{G}(\Phi^k) = [1 - F(x_{\Phi^k}^R), 1 - F(x_{\Phi^k}^L)] \\ &= \left[e^{-0.4 \cdot \left\{ 6 - \left(k - \frac{1}{2}\right) / K \right\}}, e^{-0.4 \cdot \left\{ 5 - \left(k - \frac{1}{2}\right) / K \right\}} \right] \end{aligned} \quad (3.5)$$

as displayed in Fig. 11c. Eq. (3.5) constitutes the outcome of Definition

Definition 2.11 (An FST extension principle: the $\Re$ case) We are given:

- (i) A real-valued continuous function $f: \Re \rightarrow \Re$;
- (ii) A fuzzy number A and its K -interval representation S_A ;

With the above, we:

- (a) Form the collection of intervals $\bar{C}_A^f$ based on **Definition 2.7**;
- (b) Rename the elements of $\bar{C}_A^f$ to define C_A^f ;
- (c) Define the fuzzified version $\hat{f}$ of f , with the membership function $\mu_{\hat{f}(A)}(y) \equiv \sum_{I \in C_A^f} Q(y, I)/K$.

Definition 2.7 (An extension principle for interval analysis: the $\Re$ case) We are given:

- (i) A real-valued continuous function $f: \Re \rightarrow \Re$;
- (ii) An interval I_A , and its discrete set representation $\tilde{I}_A \equiv \{x_{I_A}^m: m = 1, 2, \dots, M\}$.

With the above, we:

- (a) Form the collection of values $C_{I_A}^f$;
- (b) Define the interval-valued version $\bar{f}$ of f , to be numerically approximated as $\bar{f}(I_A) \equiv [\inf_{v \in C_{I_A}^f} \{v\}, \sup_{v \in C_{I_A}^f} \{v\}]$.

Fig. 12. Main flow of the solution process for Problem 3.1. The condensed versions of **Definition 2.11** and **Definition 2.7** are shown. **Definition 2.11** is used to solve Problem 3.1, with its step (a) relying on **Definition 2.7**.

2.7. In turn, applying Eq. (3.5) to each $k \in \{1, \dots, K\}$ yields the collection $C_\Phi^G \equiv \{\bar{G}_k | k = 1, \dots, K\}$, the desired outcome of parts (a) and (b) of **Definition 2.11**.

Finally, based on part (c) of **Definition 2.11**, we consolidate this collection of intervals to form the corresponding fuzzy-valued probability via the K -intervals representation. The membership function $\mu_{\hat{G}(\Phi)}$ of the fuzzy-valued probability $\hat{G}(\Phi)$ that the piece of manufacturing equipment has a time to failure of longer than “about 5 years” is obtained by determining the percentage of the DMs whose $\bar{G}_k$ intervals contain a given crisp value $p \in [0, 1]$. Calculations based on Eq. (2.37) in **Definition 2.11** with $K \rightarrow \infty$ give us the following analytic result as the final answer to our Problem 3.1:

$$\mu_{\hat{G}(\Phi)}(p) = \begin{cases} 6 + 2.5 \cdot \log(p) & \text{if } e^{-6 \cdot (0.4)} \leq p < e^{-5 \cdot (0.4)} \\ 1 - \{5 + (2.5 \cdot \log(p))\} & \text{if } e^{-5 \cdot (0.4)} \leq p < e^{-4 \cdot (0.4)} \\ 0 & \text{otherwise} \end{cases} \quad (3.6)$$

visualized in **Fig. 11d**. We observe that $\mu_{\hat{G}(\Phi)}$ peaks at $p = e^{-5 \cdot (0.4)} \approx 0.135$, which happens to correspond to the crisp probability that the equipment will have a time to failure of longer than 5 years (i.e., $1 - F(5) \approx 0.135$). Meanwhile, the support of $\mu_{\hat{G}(\Phi)}$ spans the interval $(e^{-6 \cdot (0.4)}, e^{-4 \cdot (0.4)}) \approx (0.091, 0.202)$, which means that the probability of the equipment having a time to failure of longer than “about 5 years” can be considered as low as 0.091 and as high as 0.202.

As a numerical check, let us also determine the probability that the equipment will have a time to failure of longer than “about 4 years” and “about 6 years”, by replacing Eq. (3.2) with

$$\mu_\Phi(x) = \begin{cases} 1 - |x - 4| & \text{if } 3 < x < 5 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\mu_{\Phi'}(x) = \begin{cases} 1 - |x - 6| & \text{if } 5 < x < 7 \\ 0 & \text{otherwise} \end{cases}$$

respectively, and then redoing the computations. This leads to the new fuzzy-valued probabilities $\hat{G}(\Phi')$ and $\hat{G}(\Phi'')$ as displayed in **Fig. 13**. Indeed, $\hat{G}(\Phi')$ and $\hat{G}(\Phi'')$ are “larger” (i.e., to the right of) and “smaller” (i.e., to the left of) than $\hat{G}(\Phi)$, respectively, confirming our intuition about the relative probabilities of these fuzzy events.

3.1.2. Example 2: Probability of fulfilling a fuzzy criterion

Before delving into the core of this example, we first need to specify some key terminologies. In the classical setting, a range of values (e.g., a temperature range) is denoted as $[T^L, T^R]$, where T^L and T^R are the real-valued left and right endpoints of the range, with $T^L < T^R$ (say $T^L = 10^\circ\text{C}$ and $T^R = 15^\circ\text{C}$). To extend the concept of “range” to the interval-mathematics setting (thus arriving at what we term an “interval-valued range”), we replace the real-valued endpoints T^L and T^R by “interval-valued endpoints” I_{T^L} and I_{T^R} (for example, $I_{T^L} = [9^\circ\text{C}, 11^\circ\text{C}]$ and $I_{T^R} = [14^\circ\text{C}, 16^\circ\text{C}]$). To further extend this concept to the fuzzy setting (thus leading to what we term a “fuzzy-valued range”), we replace I_{T^L} and I_{T^R} by “fuzzy-valued endpoints” Γ_L and Γ_R (i.e., Γ_L and Γ_R being fuzzy numbers). “Interval-valued range” and “fuzzy-valued range” enable the modeling of progressively larger degrees of uncertainties for the “range” notion.

Now, consider this scenario: Suppose that an outdoor temperature (in Celsius), when it is within a fuzzy-valued range characterized by endpoints A_L and A_R , is considered ideal for the growth of a certain tropical fruit type. The membership functions of A_L and A_R are displayed in **Fig. 14a**:

$$\mu_{A_L}(T) = \begin{cases} 1 - |T - 27| & \text{when } 26 \leq T \leq 28 \\ 0 & \text{otherwise} \end{cases} \quad (3.7.1)$$

$$\text{and } \mu_{A_R}(T) = \begin{cases} 1 - |T - 32| & \text{when } 31 \leq T \leq 33 \\ 0 & \text{otherwise} \end{cases} \quad (3.7.2)$$

It is also given that a location scouted by an agriculture firm has the following probability density distribution of daily average temperature

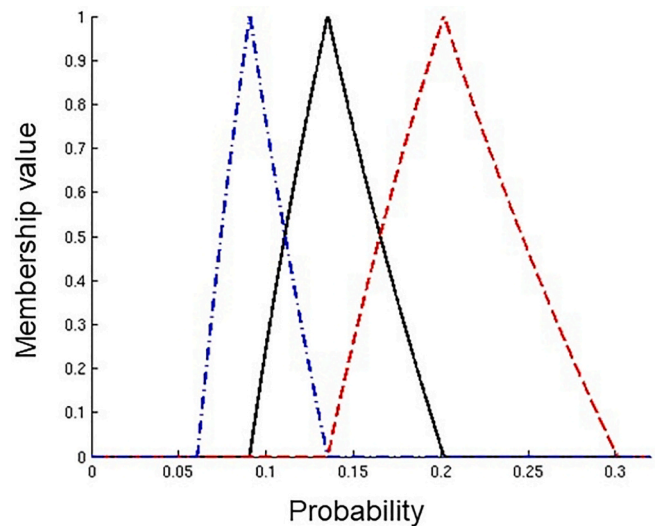


Fig. 13. The membership function of the fuzzy-valued probability $\hat{G}(\Phi)$ that the piece of equipment will have a lifetime longer than “about 4 years” (dashed line), and that of the fuzzy-valued probability $\hat{G}(\Phi')$ of a lifetime longer than about “6 years” (dash-dotted line). (Solid line: The membership function shown in **Fig. 11d**).

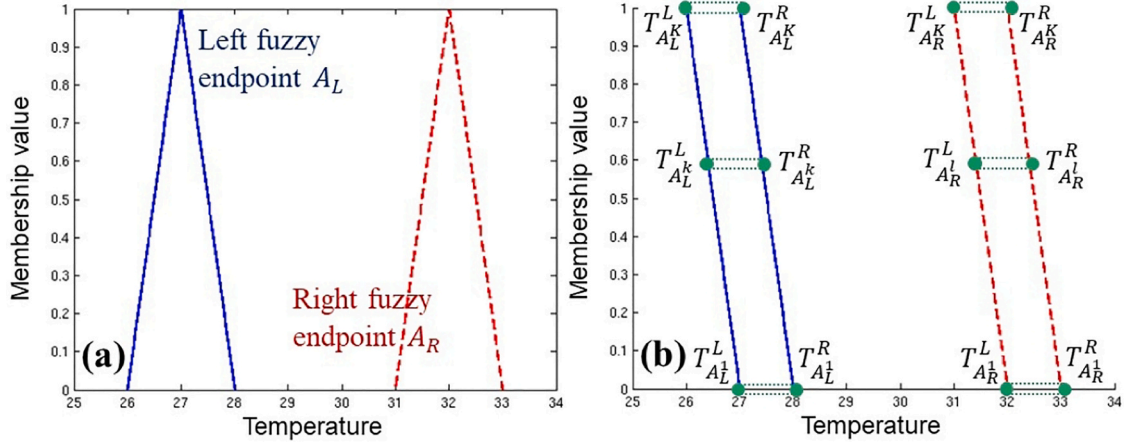


Fig. 14. Calculation of the probability of a daily average temperature falling within a fuzzy-valued range. (a) The fuzzy numbers A_L and A_R are the fuzzy-valued endpoints of the fuzzy-valued range. (b) The K -intervals representations of A_L and A_R .

during the summer season:

$$p(T) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2}\left(\frac{T - \hat{T}}{\sigma}\right)^2\right\} \quad (3.8)$$

where $\hat{T} = 31.2^\circ\text{C}$ and $\sigma = 2.5^\circ\text{C}$, hence with the associated probability function

$$P(T^L, T^R) = \int_{T=T^L}^{T=T^R} p(T) dT \quad (3.9)$$

for any real values T^L and T^R . Our problem (denoted as Problem 3.2) is: For any given day during the summer season, what is the probability that the average temperature of that day will fall within the aforementioned fuzzy-valued range? That is, what is the corresponding fuzzy-valued probability $\hat{P}(A_L, A_R)$ (in which $\hat{P}$, A_L and A_R are the fuzzified versions of P , T^L and T^R , respectively)? To solve this task, we now apply the FST extension principle for the $\mathfrak{R}^2$ case (i.e., Definition 2.12). Readers are invited to refer to Fig. 15 to view the main flow of our solution process for this problem, and to refer to Definition 2.12 in Section 2.3.3 for additional details.

Note that the term f in Definition 2.12 corresponds to the function P

in this problem. Thus, condition (i) of Definition 2.12 has been fulfilled. As indicated in condition (ii) of Definition 2.12, the next step for solving this problem involves converting each of the fuzzy numbers A_L and A_R (corresponding to the terms A_1 and A_2 in Definition 2.12) into their respective K -intervals representations. As a result, the intervals for A_L and A_R are:

$$A_L^k = [T_{A_L^k}^L, T_{A_L^k}^R] = \left[27 - \left(k - \frac{1}{2}\right)/K, 28 - \left(k - \frac{1}{2}\right)/K\right] \quad (3.10.1)$$

$$\text{and } A_R^l = [T_{A_R^l}^L, T_{A_R^l}^R] = \left[32 - \left(l - \frac{1}{2}\right)/K, 33 - \left(l - \frac{1}{2}\right)/K\right] \quad (3.10.2)$$

with $k, l \in \{1, \dots, K\}$, as displayed in Fig. 14b.

Further, condition (iii) of Definition 2.12 requires us to specify a matching pattern

$$\Pi_{A_L, A_R} \equiv \{a_{A_L, A_R}^{kl} | k, l \in \{1, \dots, K\}\} \quad (3.11)$$

in accordance with Definition 2.5' and Guideline 2.1' in Section 2.3.2.1 (also refer to the "supplemental note" following Guideline 2.1'): Given that DM k has chosen A_L^k to designate his view of A_L , what are the possible intervals from $S_{A_R} = \{A_R^1, A_R^2, \dots, A_R^K\}$ that he may possibly

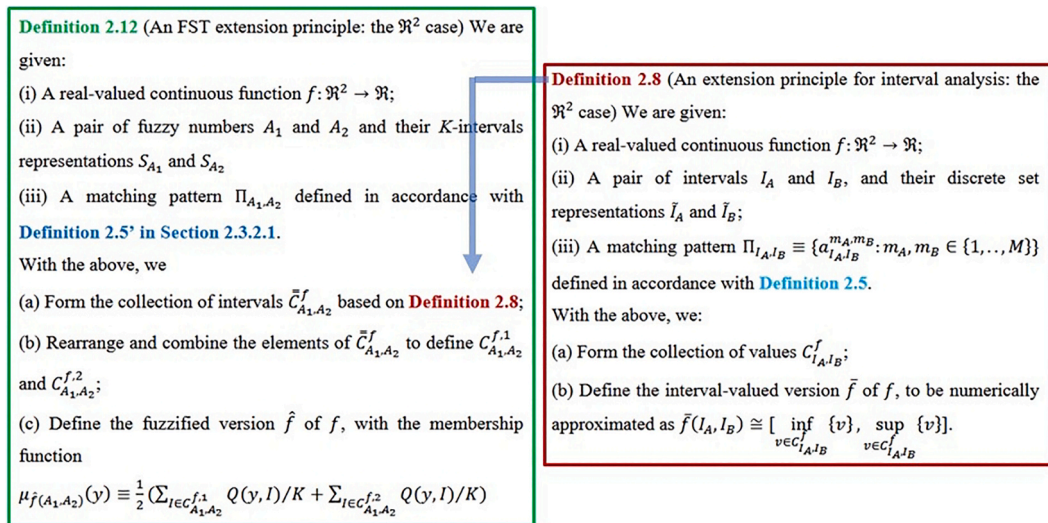


Fig. 15. Main flow of the solution process for Problem 3.2. The condensed versions of Definition 2.12 and Definition 2.8 are shown. Definition 2.12 is used to solve Problem 3.2, with its step (a) relying on Definition 2.8.

designate as his view regarding A_R ? Suppose the uncorrelated matching pattern (characterized by Eq. (2.32)) is deemed an appropriate model for the present problem. (That is, the fact that A_L^k has been picked to represent A_L does not offer sufficient information about which element (s) from S_{A_R} could be correspondingly chosen.) This leads to

$$a_{A_L A_R}^{k,l} = 1 \text{ for all } k, l \in \{1, \dots, K\} \quad (3.12)$$

as the outcome of condition (iii) of Definition 2.12.

Subsequently, we execute part (a) of Definition 2.12. In terms of the notations used in our current problem and in light of Eqs. (3.11) and (3.12), $\tilde{C}_{A_1 A_2}^f$ in Eq. (2.38) of Definition 2.12 can be written as follows:

$$\tilde{C}_{A_L A_R}^P = \{\bar{P}(A_L^k, A_R^l) | k_1, k_2 \in \{1, \dots, K\}\} \quad (3.13)$$

To finish off part (a) of Definition 2.12, we need to determine the interval-valued probability $\bar{P}(A_L^k, A_R^l)$ via Definition 2.8. In executing Definition 2.8, note that the terms f , I_A and I_B in Definition 2.8 correspond to the function P and the intervals A_L^k and A_R^l in the current problem. Thus, conditions (i) and (ii) of Definition 2.8 have been fulfilled. Condition (iii) of Definition 2.8 requires us to specify a matching pattern for the intervals A_L^k and A_R^l , i.e.,

$$\Pi_{A_L^k A_R^l} \equiv \{a_{A_L^k A_R^l}^{m_L, m_R} | m_L, m_R \in \{1, \dots, M\}\} \quad (3.14)$$

in accordance with Definition 2.5 and Guideline 2.1 in Section 2.2.1 (also refer to the “supplemental note” following Guideline 2.1): Suppose the uncorrelated matching pattern (characterized by Eq. (2.9)) is deemed an appropriate model for the present problem. (That is, the fact that a particular $T \in A_L^k = [T_{A_L^k}^L, T_{A_L^k}^R]$ has been selected does not offer sufficient information about what real value(s) from $A_R^l = [T_{A_R^l}^L, T_{A_R^l}^R]$ could be correspondingly chosen.) This leads to:

$$a_{A_L^k A_R^l}^{m_L, m_R} = 1 \text{ for all } m_L, m_R \in \{1, \dots, M\} \quad (3.15)$$

fulfilling condition (iii) of Definition 2.8. With conditions (i-iii) of Definition 2.8 satisfied, we proceed to implement part (a) of Definition 2.8. In terms of the notations used in our current problem and in light of Eqs. (3.14) and (3.15), $\tilde{C}_{I_A I_B}^f$ in Eq. (2.18) of Definition 2.8 can be written as follows:

$$C_{A_L^k A_R^l}^P \equiv \{P(T_{A_L^k}^{m_L}, T_{A_R^l}^{m_R}) | m_L, m_R \in \{1, \dots, M\}\} \quad (3.16)$$

where $T_{A_L^k}^{m_L} = (1-\lambda) \cdot T_{A_L^k}^L + \lambda \cdot T_{A_L^k}^R$ with $\lambda = \frac{m_L - \frac{1}{2}}{M - \frac{1}{2}}$, and $T_{A_R^l}^{m_R}$ is similarly defined. Subsequently, we complete part (b) of Definition 2.8, leading to the interval $\bar{P}(A_L^k, A_R^l)$ (corresponding to the term $\tilde{f}(I_A, I_B)$ in Definition 2.8) to be numerically approximated as:

$$\bar{P}(A_L^k, A_R^l) \equiv \left[\inf_{v \in C_{A_L^k A_R^l}^P} \{v\}, \sup_{v \in C_{A_L^k A_R^l}^P} \{v\} \right] \quad (3.17)$$

In fact, for the present problem, Eq. (3.17) can be determined analytically by combining Eqs. (3.7.1), (3.7.2), (3.9) and (3.10) and letting $M \rightarrow \infty$. This leads to:

$$\bar{P}(A_L^k, A_R^l) = \left[\int_{T=T_{A_L^k}^R}^{T=T_{A_R^l}^L} p(T) dT, \int_{T=T_{A_L^k}^L}^{T=T_{A_R^l}^R} p(T) dT \right] \quad (3.18)$$

Eq. (3.18) constitutes the outcome from Definition 2.8. In turn, applying Eq. (3.18) to each $k, l \in \{1, \dots, K\}$ yields the collection $\tilde{C}_{A_L A_R}^P = \{\bar{P}(A_L^k, A_R^l) | k, l \in \{1, \dots, K\}\}$, the desired outcome of part (a) of Definition

2.12.

Next, we execute part (b) of Definition 2.12 by rearranging and combining the elements of $\tilde{C}_{A_L A_R}^P$: In terms of the notations used in our current problem, $\tilde{C}_{A_1 A_2}^{f,1}$ in Eq. (2.39) and $\tilde{C}_{A_1 A_2}^{f,2}$ in Eq. (2.40) of Definition 2.12 can be written as:

$$C_{A_L A_R}^{P,1} \equiv \{\bar{P}_k | k \in \{1, \dots, K\}\} \quad (3.19)$$

$$\text{where } \bar{P}_k \equiv \bigcup \left\{ \bar{P}(A_L^k, A_R^l) | \forall l \in \{1, \dots, K\} \text{ such that } \bar{P}(A_L^k, A_R^l) \in \tilde{C}_{A_L A_R}^P \right\}$$

$$\text{and } C_{A_L A_R}^{P,2} \equiv \{\bar{P}_l | l \in \{1, \dots, K\}\} \quad (3.20)$$

$$\text{where } \bar{P}_l \equiv \bigcup \left\{ \bar{P}(A_L^k, A_R^l) | \forall k \in \{1, \dots, K\} \text{ such that } \bar{P}(A_L^k, A_R^l) \in \tilde{C}_{A_L A_R}^P \right\}$$

Finally, based on part (c) of Definition 2.12, we consolidate the two collections of intervals (i.e., $C_{A_L A_R}^{P,1}$ and $C_{A_L A_R}^{P,2}$) to form the corresponding fuzzy-valued probability via the K -intervals representation. Adapting $\mu_{\tilde{f}(A_1, A_2)}(y)$ in Eq. (2.41) to our current problem, the membership function $\mu_{\tilde{P}(A_L, A_R)}$ of the fuzzy-valued probability $\tilde{P}(A_L, A_R)$ is:

$$\mu_{\tilde{P}(A_L, A_R)}(x) \equiv \frac{1}{2} \left(\sum_{I \in C_{A_L A_R}^{P,1}} Q(x, I) / K + \sum_{I \in C_{A_L A_R}^{P,2}} Q(x, I) / K \right) \text{ where } x \in [0, 1] \quad (3.21)$$

Based on Eqs. (3.7)–(3.9) and (3.18–3.20) and letting $K \rightarrow \infty$, this can be determined analytically as:

$$\mu_{\tilde{P}(A_L, A_R)}(x) = \int_{T^*=26}^{T^*=27} Q(x, I(T^*)) dT^* \quad (3.22)$$

where

$$I(T^*) \equiv \left[\int_{T=T^*+1}^{T=31} p(T) dT, \int_{T=T^*}^{T=33} p(T) dT \right]$$

$$\text{and } Q(x, I(T^*)) = \begin{cases} 1 & \text{if } x \in I(T^*) \\ 0 & \text{otherwise} \end{cases}$$

Eq. (3.22) constitutes the final answer to our Problem 3.2, giving us the fuzzy-valued probability that a daily average temperature falls within the fuzzy-valued range with endpoints indicated by Eqs. (3.7.1) and (3.7.2). The membership function $\mu_{\tilde{P}(A_L, A_R)}$ is displayed in Fig. 16a. We observe that $\mu_{\tilde{P}(A_L, A_R)}$ reaches 1 when $x \in [0.422, 0.718]$. Meanwhile, the support of $\mu_{\tilde{P}(A_L, A_R)}$ spans the interval (0.368, 0.746).

As a numerical check, according to Eq. (3.9), the classical probability that a daily average temperature falls within the classical range $[T^L, T^R]$ with $T^L = 27^\circ\text{C}$ (corresponding to $\mu_{A_L}(T^L) = 1$ based on Eq. (3.7.1)) and $T^R = 32^\circ\text{C}$ (corresponding to $\mu_{A_R}(T^R) = 1$ based on Eq. (3.7.2)) is 0.579, which is well within the range [0.422, 0.718]. As additional numerical checks, if we make the fuzzy-valued interval larger and smaller by replacing Eqs. (3.7.1) and (3.7.2) with:

$$\mu_{A_L'}(T) = \begin{cases} 1 - |T - 26| & \text{when } 25 \leq T \leq 27 \\ 0 & \text{otherwise} \end{cases} \text{ and } \mu_{A_R'}(T) = \begin{cases} 1 - |T - 33| & \text{when } 32 \leq T \leq 34 \\ 0 & \text{otherwise} \end{cases}$$

and with:

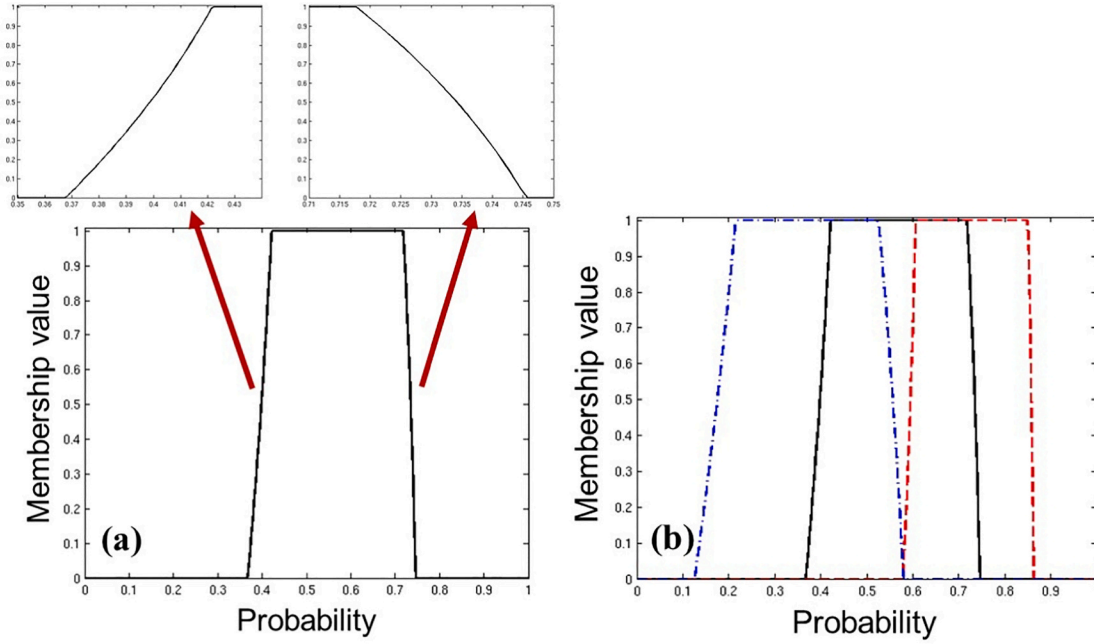


Fig. 16. (a) The membership function $\mu_{\hat{P}(A_L, A_R)}$ of the resultant fuzzy-valued probability $\hat{P}(A_L, A_R)$. The “zoom-in” version of the left edge of the membership function is shown in the upper-left sub-figure, while that of the right edge is shown in the upper-right sub-figure. (b) The membership function (dashed line) of the fuzzy-valued probability $\hat{P}(A'_L, A'_R)$ if the fuzzy-valued endpoints were A'_L and A'_R , and the membership function (dash-dotted line) of the fuzzy-valued probability $\hat{P}(A''_L, A''_R)$ if the fuzzy-valued endpoints were A''_L and A''_R .

$$\mu_{A_L''}(T) = \begin{cases} 1 - |T - 28| & \text{when } 27 \leq T \leq 29 \\ 0 & \text{otherwise} \end{cases} \text{ and } \mu_{A_R''}(T) = \begin{cases} 1 - |T - 31| & \text{when } 30 \leq T \leq 32 \\ 0 & \text{otherwise} \end{cases}$$

respectively, and then re-do the computations, we obtain the new fuzzy-valued probabilities $\hat{P}(A'_L, A'_R)$ and $\hat{P}(A''_L, A''_R)$ as displayed in Fig. 16b. Indeed, $\hat{P}(A'_L, A'_R)$ and $\hat{P}(A''_L, A''_R)$ are “larger” (i.e., to the right of) and “smaller” (i.e., to the left of) than $\hat{P}(A_L, A_R)$, respectively, confirming our intuition about the relative probabilities of these fuzzy events,

In obtaining our solution $\hat{P}(A_L, A_R)$ (shown in Fig. 16a), the uncorrelated matching pattern at the DM level (Eq. (3.12)) and at the mind-frame level (Eq. (3.15)) was used, as it was deemed that there was a lack of sufficient information to warrant stronger assumptions. Had we mistakenly chosen the perfectly correlated matching pattern for both the DM and mind-frame levels, causing a mismatch between the chosen specification and the actual situation, the resulting calculation would have produced the membership function (the dash-dotted line) displayed in Fig. 17, misleading us into a result that is much sharper than warranted. Meanwhile, mistakenly choosing the perfectly correlated matching pattern for the DM level while choosing the uncorrelated matching pattern for the mind-frame level would lead to the dashed line in Fig. 17, which is less sharp than the dash-dotted line, but still sharper than warranted. Conversely, if a situation warrants stronger assumptions but a weaker specification is used, the result may become too imprecise than necessary (as will be shown in an example in the following section). These examples show that the availability of mechanisms for selecting suitable matching patterns is essential for allowing us to create a fuzzified function that is adaptive to the assumptions of a particular application.

3.2. Correlation between two sets of fuzzy numbers in an education study

In classical statistics, given two sets of N crisp numbers $\{x_1, \dots, x_N\}$ and $\{y_1, \dots, y_N\}$, the correlation coefficient between the x 's and the y 's is

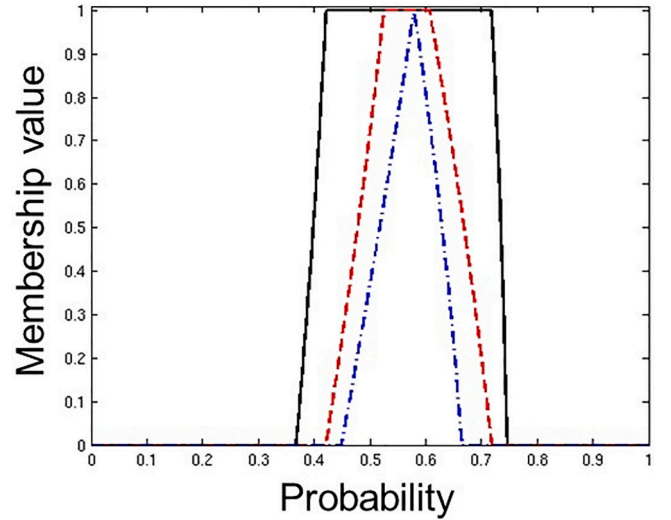


Fig. 17. Dashed line: The membership function of the resultant fuzzy-valued probability $\hat{P}(A_L, A_R)$ if perfectly correlated matching and uncorrelated matching were chosen for the DM level and the mind-frame level, respectively. Dash-dotted line: The membership function of the resultant fuzzy-valued probability if perfectly correlated matching were chosen for both the DM level and the mind-frame level. (Solid line: The membership function shown in Fig. 16a).

determined by

$$\Psi(x_1, \dots, x_N, y_1, \dots, y_N) \equiv \frac{\sum_{j=1}^N (x_j - \bar{x}) \cdot (y_j - \bar{y})}{\sqrt{\frac{1}{N-1} \sum_{j=1}^N (x_j - \bar{x})^2} \cdot \sqrt{\frac{1}{N-1} \sum_{j=1}^N (y_j - \bar{y})^2}} \quad (3.23)$$

where $\bar{x}$ and $\bar{y}$ are the average values of the x_n 's and the y_n 's, respectively. In the fuzzy setting, a natural question arises: Given two sets of N

fuzzy numbers $\{A_1, \dots, A_N\}$ and $\{B_1, \dots, B_N\}$, what is the corresponding correlation coefficient between the two sets?

To put this question into a concrete context, consider the following example: Ten college students were enrolled in an education study. Each student wrote two academic essays of different topics, one in English and the other in Chinese. A panel of teachers (DMs) evaluated the English and Chinese essays of each student. The aggregation of these assessments resulted in two fuzzy numbers for each student: one measuring his/her English composition ability and the other measuring his/her Chinese composition ability. The resultant data, $A_1, \dots, A_{10}$ and $B_1, \dots, B_{10}$, is presented in Table 1.

As part of this study, the researchers would like to quantify the correlation between the English and Chinese composition abilities of these students, i.e., $\hat{\Psi}(A_1, \dots, A_{10}, B_1, \dots, B_{10})$ (denoted as Problem 3.3). Here, $\hat{\Psi}$, A_n and B_n are the fuzzified versions of Ψ , x_n and y_n in Eq. (3.23), respectively. To find the fuzzy-valued correlation coefficient $\hat{\Psi}(A_1, \dots, A_{10}, B_1, \dots, B_{10})$, we now apply the FST extension principle for the $\mathfrak{N}^N$ case (i.e., Definition 2.13). Readers are invited to refer to Fig. 18 to view the main flow of our solution process for this problem, and to refer to Definition 2.13 in Section 2.3.3 for additional details.

By applying Definition 2.2 to the data given in Table 1, the numerical values for $x_{A_n}^L$ and $x_{A_n}^R$ for all $n \in \{1, \dots, 20\}$ and all $k_n \in \{1, \dots, K\}$ can be obtained (not shown to conserve space).

Further, condition (iii) of Definition 2.13 requires us to choose a full specification of matching patterns Y for the set of fuzzy numbers $A_1, \dots, A_{20}$ in accordance with Definitions 2.6' and Guideline 2.1' in Section 2.3.2.1 (also refer to the "supplemental note" following Guideline 2.1'): For example, given that DM k has chosen the interval A_1^k to designate his view of A_1 (i.e., his view about the English composition ability of Student 1), what are the possible intervals from $S_{A_n} = \{A_n^1, A_n^2, \dots, A_n^K\}$ that he may possibly designate as his view regarding A_n ? Suppose it is observed that the DMs can be approximately categorized into one of the three groups: more stringent, moderately stringent, and less stringent. Then, the matching pattern $\Pi_{A_n, A_{n+1}}$ for A_n and A_{n+1} (with $n \in \{1, \dots, 19\}$) depicted in Fig. 19 can be deemed appropriate to capture this observation. In equation form, this matching pattern can be written as:

$$\Pi_{A_n, A_{n+1}} \equiv \left\{ a_{A_n, A_{n+1}}^{k_n, k_{n+1}} \mid k_n, k_{n+1} \in \{1, \dots, K\} \right\} \text{ for } n = 1, \dots, 19 \quad (3.26)$$

where

$$a_{A_n, A_{n+1}}^{k_n, k_{n+1}} = \begin{cases} 1 & \text{if } 1 \leq k_n, k_{n+1} \leq \frac{K}{3} \text{ or } \left(\frac{K}{3} + 1\right) \leq k_n, k_{n+1} \leq \frac{2K}{3} \text{ or } \left(\frac{2K}{3} + 1\right) \leq k_n, k_{n+1} \leq K \\ 0 & \text{otherwise} \end{cases}$$

Note that the function Ψ in the present problem corresponds to f in Definition 2.13. Thus, condition (i) of Definition 2.13 has been fulfilled. Meanwhile, as indicated above, the fuzzy numbers of interests in the present problem are $A_1, \dots, A_{10}, B_1, \dots, B_{10}$, but to simplify notations for our subsequent discussion, let's rename $B_1, \dots, B_{10}$ as $A_{11}, \dots, A_{20}$. This means that the set of fuzzy numbers we deal with are labelled as $A_1, \dots, A_{20}$.

Now, as indicated in condition (ii) of Definition 2.13, the next step in solving this problem involves converting each of the fuzzy numbers A_n into their respective K -intervals representations, i.e.,

$$S_{A_n} = \{A_n^{k_n} \mid k_n = 1, \dots, K\} \text{ for } n = 1, \dots, 20 \quad (3.24)$$

where an interval $A_n^{k_n}$ can be denoted as

$$A_n^{k_n} = [x_{A_n}^L, x_{A_n}^R] \quad (3.25)$$

Table 1

Ratings of English and Chinese composition abilities of ten college students. For instance, the English composition ability of Student 10 is represented by the triangular fuzzy number (56, 63, 68). The associated membership function of the fuzzy number is defined as follows: .

$$\mu_{Eng_{10}}(x) = \begin{cases} (x - 56)/(63 - 56) & \text{if } 56 < x < 63 \\ (68 - x)/(68 - 63) & \text{if } 63 < x < 68 \\ 0 & \text{otherwise} \end{cases}$$

Student number	English composition rating	Chinese composition rating
1	(80, 85, 90)	(74, 79, 83)
2	(50, 60, 65)	(63, 66, 70)
3	(90, 95, 100)	(74, 78, 83)
4	(42, 50, 54)	(70, 78, 83)
5	(95, 97, 100)	(86, 93, 97)
6	(30, 35, 41)	(60, 67, 73)
7	(82, 87, 92)	(65, 72, 78)
8	(56, 62, 67)	(78, 83, 86)
9	(77, 84, 89)	(64, 69, 74)
10	(56, 63, 68)	(82, 87, 90)

Eq. (3.26) yields a full specification of matching patterns Y for the fuzzy numbers $A_1, \dots, A_{20}$. Thus, condition (iii) of Definition 2.13 is fulfilled.

Subsequently, we execute part (a) of Definition 2.13. In terms of the notations used in our current problem, $\tilde{C}_{A_1, \dots, A_n}^f$ in Eq. (2.42) of Definition 2.13 can be written as follows:

$$\tilde{C}_{A_1, \dots, A_{20}}^\Psi = \tilde{C}_{A_1, \dots, A_{20}, 1}^\Psi \cup \tilde{C}_{A_1, \dots, A_{20}, 2}^\Psi \cup \tilde{C}_{A_1, \dots, A_{20}, 3}^\Psi \quad (3.27.1)$$

$$\text{where } \tilde{C}_{A_1, \dots, A_{20}, 1}^\Psi = \left\{ \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \mid 1 \leq k_n, k_{n+1} \leq \frac{K}{3} \text{ for } n = 1, \dots, 19 \right\} \quad (3.27.2)$$

$$\tilde{C}_{A_1, \dots, A_{20}, 2}^\Psi = \left\{ \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \mid \left(\frac{K}{3} + 1\right) \leq k_n, k_{n+1} \leq \frac{2K}{3} \text{ for } n = 1, \dots, 19 \right\} \quad (3.27.3)$$

$$\text{and } \tilde{C}_{A_1, \dots, A_{20}, 3}^\Psi = \left\{ \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \mid \left(\frac{2K}{3} + 1\right) \leq k_n, k_{n+1} \leq K \text{ for } n = 1, \dots, 19 \right\} \quad (3.27.4)$$

To complete part (a) of Definition 2.13, we need to determine the interval-valued correlation coefficient $\bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}})$ via Definition 2.9.

Note that the terms f and I_n (for $n \in \{1, \dots, N\}$) in Definition 2.9 correspond to the function Ψ and the intervals $A_n^{k_n}$ (for $n \in \{1, \dots, 20\}$; refer to Eq. (3.25)) in the current problem. Thus, conditions (i) and (ii) of Definition 2.9 have been fulfilled. Condition (iii) of Definition 2.9 requires us to specify the matching pattern for each pair of intervals $A_n^{k_n}$ and $A_{n+1}^{k_{n+1}}$, i.e.,

$$\Pi_{A_n, A_{n+1}}^{k_n, k_{n+1}} \equiv \left\{ a_{A_n, A_{n+1}}^{m_n, m_{n+1}} : m_n, m_{n+1} \in \{1, \dots, M\} \right\} \text{ with } n = 1, \dots, 19 \quad (3.28)$$

Definition 2.13 (An FST extension principle: the $\mathfrak{R}^N$ case) We are given:

- (i) A real-valued continuous function $f: \mathfrak{R}^N \rightarrow \mathfrak{R}$;
- (ii) A set of N fuzzy numbers $\Omega \equiv \{A_1, \dots, A_N\}$ and their K -intervals representations $S_{A_1}, \dots, S_{A_N}$;
- (iii) A full specification of matching patterns Υ for the set of fuzzy numbers $\Omega \equiv \{A_1, \dots, A_N\}$ defined in accordance with **Definitions 2.6*** in **Section 2.3.2.1**.

With the above, we:

- (a) Form the collection of intervals $\bar{C}_{A_1 \dots A_N}^f$ based on **Definition 2.9**;
- (b) Rearrange and combine the elements of $\bar{C}_{A_1 \dots A_N}^f$ to define $C_{A_1 \dots A_N}^{f,1}, C_{A_1 \dots A_N}^{f,2}, \dots, C_{A_1 \dots A_N}^{f,N}$;
- (c) Define the fuzzified version $\hat{f}$ of f , with the membership function

$$\mu_{\hat{f}(A_1 \dots A_N)}(y) \equiv \left(\sum_{I \in C_{A_1 \dots A_N}^{f,1}} Q(y, I)/K + \dots + \sum_{I \in C_{A_1 \dots A_N}^{f,N}} Q(y, I)/K \right) / N$$

Definition 2.9 (An extension principle for interval analysis: the $\mathfrak{R}^N$ case) We are given:

- (i) A real-valued continuous function $f: \mathfrak{R}^N \rightarrow \mathfrak{R}$;
- (ii) A set of intervals $S = \{I_1, \dots, I_N\}$, and their discrete set representations $\tilde{I}_1, \dots, \tilde{I}_N$;
- (iii) A full specification of matching patterns Λ for the set of intervals $S = \{I_1, \dots, I_N\}$ in accordance with **Definition 2.6**.

With the above, we:

- (a) Form the collection of values $C_{I_1 \dots I_N}^f$;
- (b) Define the interval-valued version $\bar{f}$ of f , to be numerically approximated as $\bar{f}(I_1, \dots, I_N) \cong [\inf_{v \in C_{I_1 \dots I_N}^f} \{v\}, \sup_{v \in C_{I_1 \dots I_N}^f} \{v\}]$.

Fig. 18. Main flow of the solution process for Problem 3.3. The condensed versions of **Definition 2.13** and **Definition 2.9** are shown. **Definition 2.13** is used to solve Problem 3.3, with its step (a) relying on **Definition 2.9**.

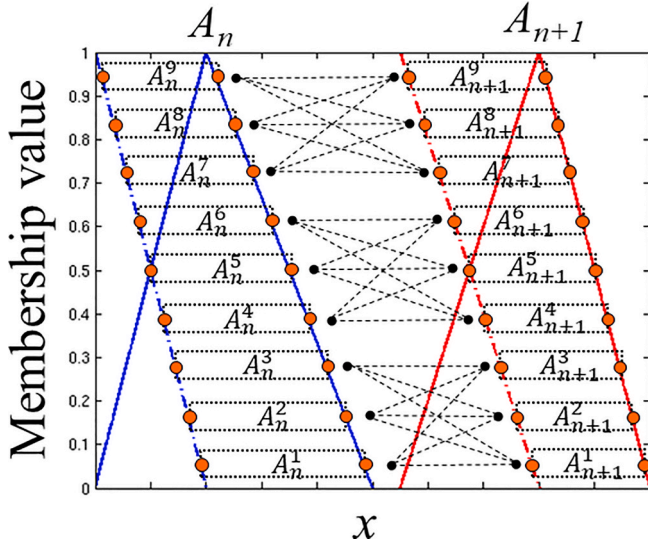


Fig. 19. The matching pattern for two sets of intervals $S_n = \{A_n^1, A_n^2, \dots, A_n^K\}$ and $S_{n+1} = \{A_{n+1}^1, A_{n+1}^2, \dots, A_{n+1}^K\}$, where S_n and S_{n+1} are the K -intervals decompositions of given fuzzy numbers A_n and A_{n+1} . For illustration purposes, let us choose $K = 9$, and focus on DMs 1, 5 and 9. DM 1 belongs to the group of less stringent DMs. The fact that he designates A_n^1 as his view for A_n means that he may possibly choose A_{n+1}^1, A_{n+1}^2 or A_{n+1}^3 to represent his view for A_{n+1} . Meanwhile, DM 5 belongs to the group of moderately stringent DMs. The fact that she designates A_n^5 as her view for A_n means that she may possibly choose A_{n+1}^4, A_{n+1}^5 or A_{n+1}^6 to represent her view for A_{n+1} . Lastly, DM 9 belongs to the group of more stringent DMs. The fact that he designates A_n^9 as his view for A_n means that he may possibly choose A_{n+1}^7, A_{n+1}^8 or A_{n+1}^9 to represent his view for A_{n+1} .

in accordance with **Definitions 2.6** and **Guideline 2.1** in **Section 2.2.1** (also refer to the “supplemental note” following **Guideline 2.1**): Suppose the perfectly correlated matching pattern (characterized by Eq. (2.8)) is deemed an appropriate model (i.e., the behavior of a DM being similar to

that of the manager described in Scenario I discussed in **Section 2.2.1**). This leads to:

$$a_{A_n^{k_n} A_{n+1}^{k_{n+1}}}^{m_n, m_{n+1}} = \begin{cases} 1 & \text{if } m_n = m_{n+1} \text{ for } m_n, m_{n+1} \in \{1, \dots, M\} \\ 0 & \text{otherwise} \end{cases} \quad (3.29)$$

The combination of Eqs. (3.28) and (3.29) yields a full specification of matching pattern for the intervals $A_n^{k_n}$ (with $n \in \{1, \dots, 20\}$). Thus, condition (iii) of **Definition 2.9** is fulfilled.

We are now ready to implement part (a) of **Definition 2.9**: In terms of the notations used in our current problem and in light of Eqs. (3.28) and (3.29), $C_{I_1 \dots I_N}^f$ in Eq. (2.20) of **Definition 2.9** can be written as follows:

$$C_{A_1^{k_1} \dots A_{20}^{k_{20}}}^\Psi \equiv \left\{ \Psi \left(x_{A_1^{k_1}}^m, \dots, x_{A_{20}^{k_{20}}}^m \right) \mid m \in \{1, \dots, M\} \right\} \quad (3.30)$$

where $x_{A_n^{k_n}}^m = (1 - \lambda) \cdot x_{A_n^{k_n}}^L + \lambda \cdot x_{A_n^{k_n}}^R$ with $\lambda = \frac{m-1}{M}$ with $x_{A_n^{k_n}}^L$ and $x_{A_n^{k_n}}^R$ being the left and right endpoints of $A_n^{k_n}$ according to Eq. (3.25). Consequently, based on part (b) of **Definition 2.9**, the interval $\bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}})$ can be numerically approximated as:

$$\bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \cong \left[\inf_{v \in C_{A_1^{k_1} \dots A_{20}^{k_{20}}}^\Psi} \{v\}, \sup_{v \in C_{A_1^{k_1} \dots A_{20}^{k_{20}}}^\Psi} \{v\} \right] \quad (3.31)$$

Eq. (3.31) constitutes the outcome from **Definition 2.9**. In turn, applying Eq. (3.31) to determine the elements of $\bar{C}_{A_1 \dots A_{20}, 1}^\Psi$, $\bar{C}_{A_1 \dots A_{20}, 2}^\Psi$ and $\bar{C}_{A_1 \dots A_{20}, 3}^\Psi$ in Eqs. (3.27.2), (3.27.3) and (3.27.4) respectively yields the collection $\bar{C}_{A_1 \dots A_{20}}^\Psi$, the desired outcome of part (a) of **Definition 2.13**.

Next, we execute part (b) of **Definition 2.13** by rearranging and combining the elements of $\bar{C}_{A_1 \dots A_{20}}^\Psi$: In terms of the notations used in our current problem, $C_{A_1 \dots A_N}^{f,1}, \dots, C_{A_1 \dots A_N}^{f,N}$ (described in Eqs. (2.43) and (2.44)) of **Definition 2.13** can be written as:

$$C_{A_1 \dots A_{20}}^{\Psi,1} \equiv \{ \bar{\Psi}_{k_1} \mid k_1 \in \{1, \dots, K\} \} \quad (3.32)$$

$$\text{with } \bar{\Psi}_{k_1} \equiv \bigcup \left\{ \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \mid \forall k_2, \dots, k_{20} \in \{1, \dots, K\} \text{ such that } \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \in \tilde{C}_{A_1, \dots, A_{20}}^{\Psi} \right\}$$

$$\text{and } C_{A_1, \dots, A_{20}}^{\Psi, 2} \equiv \{ \bar{\Psi}_{k_2} \mid k_2 \in \{1, \dots, K\} \} \quad (3.33)$$

$$\text{with } \bar{\Psi}_{k_2} \equiv \bigcup \left\{ \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \mid \forall k_1, k_3, k_4, \dots, k_{20} \in \{1, \dots, K\} \text{ such that } \bar{\Psi}(A_1^{k_1}, \dots, A_{20}^{k_{20}}) \in \tilde{C}_{A_1, \dots, A_{20}}^{\Psi} \right\} \text{ and so on.}$$

Finally, based on part (c) of Definition 2.13, we consolidate the N collections $C_{A_1, \dots, A_{20}}^{\Psi, 1}, \dots, C_{A_1, \dots, A_{20}}^{\Psi, 20}$ of intervals to form the corresponding fuzzy-valued correlation coefficient via the K -intervals representation. Adapting $\mu_{\hat{f}(A_1, \dots, A_N)}(y)$ in Eq. (2.45) to our current problem, the membership function $\mu_{\hat{\Psi}(A_1, \dots, A_{20})}$ of the fuzzy-valued correlation coefficient $\hat{\Psi}(A_1, \dots, A_{20})$ is:

$$\mu_{\hat{\Psi}(A_1, \dots, A_{20})}(z) \equiv \left(\sum_{I \in C_{A_1, \dots, A_{20}}^{\Psi, 1}} Q(z, I) / K + \sum_{I \in C_{A_1, \dots, A_{20}}^{\Psi, 2}} Q(z, I) / K + \dots + \sum_{I \in C_{A_1, \dots, A_{20}}^{\Psi, 20}} Q(z, I) / K \right) / 20 \quad (3.34)$$

where $z \in [0, 1]$. By carrying out numerical calculations on Eqs. (3.27), (3.30), (3.31) and (3.34), $\mu_{\hat{\Psi}(A_1, \dots, A_{20})}$ is obtained and displayed in Fig. 20a. Eq. (3.34) together with this numerical result constitutes the final answer to our Problem 3.3, giving us the fuzzy coefficient that quantifies the correlation between the English and Chinese composition abilities of the ten students.

As a numerical check, based on Eq. (3.23) and Table 1, we calculate the classical correlation coefficient for $\{x_{A_1}^*, \dots, x_{A_{10}}^*\}$ versus $\{y_{B_1}^*, \dots, y_{B_{10}}^*\}$, with $x_{A_n}^*$ and $y_{B_n}^*$ defined as the locations associated with the peaks of the membership functions μ_{A_n} and μ_{B_n} for the fuzzy numbers A_n and B_n , respectively (e.g., $x_{A_{10}}^* = 63$ and $y_{B_{10}}^* = 87$). This results in:

$$\Psi(x_{A_1}^*, \dots, x_{A_{10}}^*, y_{B_1}^*, \dots, y_{B_{10}}^*) = 0.368 \quad (3.35)$$

which is well within the interval $\{z \in \mathbb{R} : \mu_{\hat{\Psi}(A_1, \dots, A_{20})}(z) = 1\}$ as indicated

in Fig. 20a.

Suppose our FST user disregarded the assumption described in Fig. 19 (i.e., that the DMs are categorized into one of the “more stringent”, “moderately stringent”, and “less stringent” groups) and instead imposed the perfectly correlated matching pattern for A_n and A_{n+1} (for $n \in \{1, \dots, 19\}$). This would result in an assumption about the relative stringencies of the DMs across the A_n ’s that is stronger than warranted. Indeed, the ensuing computation based on this assumption would have misled our user into a result that is too sharp (the dash-dotted line depicted in Fig. 20b). Conversely, suppose our user disregarded the assumption described in Fig. 19 and instead imposed the uncorrelated matching pattern for A_n and A_{n+1} (for $n \in \{1, \dots, 19\}$). This would result in an assumption that is weaker than necessary. The subsequent computation based on it would have misled our user into a result that is too imprecise (the dashed line depicted in Fig. 20b).

These examples again show that the availability of mechanisms for prescribing appropriate matching patterns is important in allowing us to develop a fuzzified function that is adaptive to the assumptions of a given application.

4. Comparisons with previous works

4.1. Comparisons with other alternative FST extension principles

In summary, a bird’s eye view of our proposed FST extension principle reveals the following building blocks. Given a real-valued function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ and a set of fuzzy numbers $A_1, \dots, A_N$ of interest, we:

- (i) **Decompose each fuzzy numbers** into its corresponding K -intervals representation (i.e., a collection of intervals).
- (ii) **Decompose each interval** from Step (i) into its discrete set representation (i.e., a collection of real values).
- (iii) **Extend the real-value function f** to become the corresponding interval-valued function $\bar{f}$. Specifically: (a) The outcomes (i.e., the real values) from Step (ii) serve as inputs to f , yielding real-valued outputs. (b) These real-valued outputs are then consolidated into corresponding intervals. These resultant intervals serve as the outputs of $\bar{f}$.
- (iv) **Extend the interval-valued function $\bar{f}$** to become the corresponding fuzzy-valued function $\hat{f}$. Specifically, the outputs of $\bar{f}$ from Step (iii) (i.e., the collection of intervals) are consolidated into a corresponding fuzzy number via the K -intervals representation.

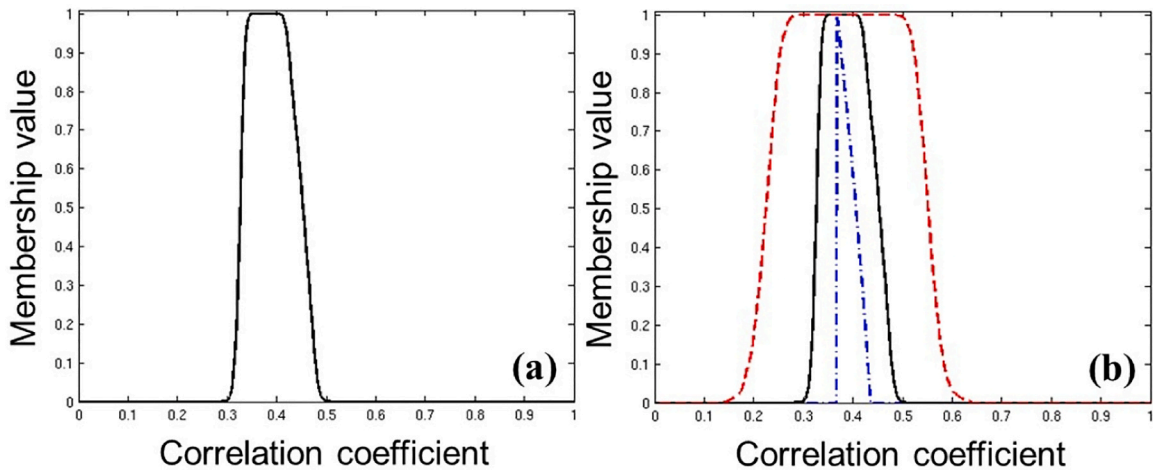


Fig. 20. Calculation of the fuzzy-valued correlation coefficient. (a) The membership function $\mu_{\hat{\Psi}(A_1, \dots, A_{20})}$ of the resultant fuzzy-valued correlation coefficient $\hat{\Psi}(A_1, \dots, A_{20})$. (b) The membership function (dash-dotted line) of the corresponding fuzzy-valued correlation coefficient if the matching pattern described in Fig. 19 were replaced by the perfectly correlated matching pattern, and the membership function (dashed line) of the corresponding fuzzy-valued correlation coefficient if the matching pattern described in Fig. 19 were replaced by the uncorrelated matching pattern.

- (v) **Control the interaction** of the real values (mentioned in Steps (ii) and (iii)) and of the intervals (mentioned in Steps (iii) and (iv)) directly by our FST user. This is done by prescribing matching patterns based on [Definition 2.5](#), Guideline 2.1 and [Definition 2.6](#) in [Section 2.2.1](#) (to control the interactivity of real values) and based on [Definition 2.5'](#), Guideline 2.1' and [Definition 2.6'](#) in [Section 2.3.2.1](#) (to control the interactivity of intervals).

Arguably, each of these steps uses concepts and operations in a manner that is well-acquainted to classically trained engineers. For example, a given fuzzy set A (a non-classical mathematical object) is simply decomposed into a set of intervals (a type of mathematical object that all engineers are familiar with) via the K -intervals representation. Consequently, the membership degree of a given element $x \in \mathfrak{N}$ with respect to A is simply determined as the percentage of those intervals that contain x . In other words, the concept of “membership degree” (a non-classical concept) is defined in terms of lower-level notions that are concrete and basic (i.e., “intervals” and “percentage”) to classically trained engineers.

In contrast, when the “membership degree” is regarded as a base concept itself and is not further defined in terms of concrete lower-level notions, it may become more challenging for engineers without FST expertise to adapt existing FST extension principles to their practical applications. (Therefore, we believe there is room for alternative frameworks that are more accessible to engineers without FST expertise.) For example, there have been questions among FST experts (let alone the non-experts) regarding the suitability of certain components in [Eq. \(1.3\)](#):

- (i) In the early days after FST was proposed, [Jain \(1976\)](#) questioned the use of the supremum operation in [Eq. \(1.3\)](#) and proposed using a probabilistic sum instead. [Dubois and Prade \(1980\)](#) suggested replacing the “min” operator in [Eq. \(1.3\)](#) by the product operator. Subsequently, [De Cooman et al. \(1990\)](#) and [Fullér and Keresztfalvi \(1990\)](#) proposed a generalization in which the “min” operator was replaced with the t -norm.
- (ii) Among the later works, [Wu \(2010, 2019\)](#) suggested generalizing the extension principle by replacing the “min” operator in [Eq. \(1.3\)](#) by an operator $W_N : [0, 1]^N \rightarrow [0, 1]$ that did not assume any additional conditions.
- (iii) In a recent work, [Islam et al. \(2021\)](#) pointed out that when mapping $A_1, \dots, A_N$ to C based on ZEP in [Definition 1.2](#), the height (i.e., the maximum membership value) of C was always restricted to be the lowest height among the input fuzzy sets $A_1, \dots, A_N$. They argued that such a restriction might not be reasonable and proposed a generalized extension principle by replacing the “min” operator in [Eq. \(1.3\)](#) with a general membership mapping function $f_N : [0, 1]^N \rightarrow [0, 1]$, which was required to satisfy the properties that f_N is non-decreasing, $f_N(0, \dots, 0) = 0$, and $f_N(1, \dots, 1) = 1$.

These and other works have certainly contributed to offering a richer collection of extension principles for the FST users, with their proposals of using t -norms, the operator W_N , and the mapping function f_N to replace the “min” operator in the original FST extension principle. However, there are two issues to consider:

- (i) Each of these replacements (e.g., the t -norm, W_N and f_N) represents a broad class of functions. Arguably, these works offer little to no concrete guidelines for non-experts on how to construct, in a step-by-step manner, specific functions (say within the class of t -norms) that can fit well with the assumptions of their applications. Our view is that providing such concrete guidelines is challenging when “membership degree” is regarded as a base concept and is not further defined in terms of concrete lower-level notions.
- (ii) All of the works discussed above (except [Jain, 1976](#)) retained the use of the supremum operator described in [Eq. \(1.3\)](#) without further explanation. We have argued in [Section 1](#) that the justification of using the supremum operator to determine membership values can be rather

opaque to FST users, especially those who may not possess deep FST expertise. In particular, as illustrated by the court trial example in that section, a supremum-based extension principle like [Definition 1.2](#) could lead to results that seemingly deviate from those expected by an FST user based on common-sense reasoning. [Appendix A](#) illustrates with a simple example how our proposed extension principle can address this issue.

Among the various alternative extension principles proposed in recent FST literature, we would like to highlight a noteworthy approach termed the sup-J extension principle for generalizing [Definition 1.2](#) (E. Esmi, Sussner, Ignácio, & de Barros, 2018; E. Esmi, Wasques, & de Barros, 2021; Fullér & Majlender, 2004; Sánchez, Wasques, Arenas, Esmi, & de Barros, 2021). Its key building block is the concept of a joint possibility distribution J of given fuzzy numbers $A_1, \dots, A_N$, which satisfies the condition that:

$$\mu_{A_n}(x) = \sup_{\{(x_1, \dots, x_N) : x_n = x\}} J(x_1, \dots, x_N) \text{ for all } x \in \mathfrak{N} \text{ and } n \in \{1, \dots, N\} \quad (4.1)$$

where μ_{A_n} is the membership function of A_n . Based upon [Eq. \(4.1\)](#), for any given real-valued function $f : \mathfrak{N}^N \rightarrow \mathfrak{N}$, the sup-J extension of f at $(A_1, \dots, A_N)$ is defined as:

$$\hat{f}_J(A_1, \dots, A_N)(y) \equiv \sup_{(x_1, \dots, x_N) \in f^{-1}(y)} J(x_1, \dots, x_N) \quad (4.2)$$

with $f^{-1}(y)$ being

$$f^{-1}(y) = \{(x_1, \dots, x_N) \in \mathfrak{N}^N \mid f(x_1, \dots, x_N) = y\}$$

The flexibility afforded by [Eq. \(4.1\)](#) allows for various types of J for the sup-J extension principle, such as:

- (i) The t -norm based joint possibility distributions ([Fullér & Keresztfalvi, 1992](#))

$$J_t(x_1, \dots, x_N) \equiv \mu_{A_1}(x_1) \Delta \mu_{A_2}(x_2) \Delta \dots \Delta \mu_{A_N}(x_N) \quad (4.3)$$

(where Δ is any given t -norm).

- (ii) The linear-interactivity-based joint possibility distributions ([Carlsson, Fullér, & Majlender, 2004](#); E. Esmi, Sánchez, Wasques, & de Barros, 2021)

$$J_L(x_1, \dots, x_N) \equiv \mu_{A_n}(x_n) \bullet \chi_L(x_1, \dots, x_N) \text{ for all } (x_1, \dots, x_N) \in \mathfrak{N}^N \quad (4.4)$$

with χ_L being the characteristic function of the line $L = \{(u, q_2 \bullet u + r_2, \dots, q_N \bullet u + r_N) \mid u \in \mathfrak{N}\}$ where $q_2, \dots, q_N$ and $r_2, \dots, r_N$ are pre-selected real values.

- (iii) Families of joint possibility distributions have also been developed in noteworthy works like E. Esmi et al. (2021), E. Esmi et al. (2018), E. Esmi et al. (2021) and [Pedro, Lopes, Wasques, Esmi, and de Barros \(2023\)](#), leading to interesting results such as interactive arithmetic.

A key advantage of the sup-J extension principle is that, since it is based on the concept of joint possibility distribution (i.e., [Eq. \(4.1\)](#)), different types of interactivity among a given set of fuzzy numbers of interest can be imposed (e.g., via [Eqs. \(4.3\) and \(4.4\)](#)). Additionally, as it is based on [Eq. \(4.1\)](#), the sup-J extension principle is deeply connected with all theoretical results from the theory of joint possibility distributions. These deep connections offer exceptional opportunities for FST experts to further develop the sup-J extension principle. However, for an FST user who does not possess deep FST expertise, the sup-J extension principle remains a *supremum-based* extension principle. Thus, non-experts using it would still encounter the $\mu_C(y^*)$ versus $\mu_C(y^{**})$ dilemma discussed in the court trial example in [Section 1](#). In summary, while developments like the sup-J extension principle have tremendous potential to flourish on the FST theoretical front, we can still address the needs of non-experts by introducing alternative yet concrete methodologies like the one we propose in this article.

4.2. Comparison with other fuzzy-valued probability and fuzzy-valued correlation measures

Hesamian and Shams (2017) provided a concise summary of the FST literature regarding the probability theory of fuzzy events. Rediscussing those and other works exhaustively here would be beyond the scope of this article, as our primary aim is to demonstrate the utility of our proposed extension principle for *generally* extending classical mathematics (not just restricted to probability theory) to fuzzy mathematics. Thus, as illustrative examples of previous works, we will discuss the landmark formulations by Zadeh (1968), Yager (1979, 1984) and Klement (1982), as well as the follow-up work by Hesamian and Shams (2017).

In classical probability theory, (X, Ξ, P) denotes a probability space, where X is a sample space, Ξ is a σ -field of Borel sets in X , and P is a real-valued function which assigns to every event $E \in \Xi$ its probability $P(E)$. For simplicity, consider the case when $X = \mathfrak{N}$. Then, given an $E \in \Xi$, the probability of E is expressed as

$$P(E) = \int_{x \in \mathfrak{N}} \psi_E(x) dP(x) \quad (4.5)$$

where

$$\psi_E(x) = \begin{cases} 1 & \text{for } x \in E \\ 0 & \text{otherwise} \end{cases}$$

Having a fuzzy event defined as a fuzzy set Θ with membership function f_Θ , Zadeh (1968) postulated the probability of the fuzzy event Θ by heuristically replacing ψ_E with f_Θ as

$$P_Z(\Theta) = \int_{x \in \mathfrak{N}} f_\Theta(x) dP(x) \quad (4.6)$$

$$\text{which is equivalent to } P_Z(\Theta) = \int_{x \in \mathfrak{N}} P(\Theta_\alpha) d\alpha \quad (4.7)$$

where Θ_α is the α -cut of Θ , i.e., $\Theta_\alpha = \{x \in \mathfrak{N} | f_\Theta(x) \geq \alpha\}$.

Yager (1979) pointed out that $P_Z(\Theta)$ always yielded a real value for a given fuzzy event Θ , and hence $P_Z(\Theta)$ (due to its lack of fuzziness) was inconsistent with FST. Via Zadeh's extension principle, Yager postulated an alternative formula:

$$P_{Y1}(\Theta)(w) = \sup_{\alpha \in [0,1]} \{\alpha | w = P(\Theta_\alpha)\} \text{ for } w \in [0, 1] \quad (4.8)$$

$P_{Y1}(\Theta)$ was interpreted by Yager as "the probability of at least an α degree of satisfaction to the condition Θ ". Subsequently, Klement (1982) proposed a heuristic modification of $P_{Y1}(\Theta)$, leading to:

$$P_K(\Theta)(w) = \sup_{\alpha \in [0,1]} \{\alpha | w \leq P(\Theta_\alpha)\} \text{ for } w \in [0, 1] \quad (4.9)$$

Yager (1984) in turn modified Klement's postulation, resulting in:

$$P_{Y2}(\Theta)(w) = \min\{P_K(\Theta)(w), P_{Y1}(\Theta)(w)\} \text{ for } w \in [0, 1] \quad (4.10)$$

where $P_K(\Theta)(w) = \sup_{\alpha \in [0,1]} \{\alpha | w \geq P(\Theta_\alpha)\}$ Yager (1984) interpreted

$P_K(\Theta)(w)$ as the degree of truth of the proposition "the probability of Θ is at least w ", and $P_{Y2}(\Theta)(w)$ as the degree of truth of the proposition "the probability of Θ is exactly w ". Following up on these landmark works, Hesamian and Shams (2017) proposed a further modified postulation:

$$P_{HS}(\Theta)(w) = \sup_{\alpha \in [0,1]} \{\alpha | P(\Theta_{1-\alpha/2}) \leq w \leq P(\Theta_{\alpha/2})\} \text{ for } w \in [0, 1] \quad (4.11)$$

The above examples illustrate the difficulties in developing a simple-to-understand probability theory of fuzzy events: (i) $P_Z(\Theta)$ led to results that are inconsistent with FST; (ii) In $P_{Y1}(\Theta)$, the supremum operation over all $\alpha \in [0, 1]$ was used. Thus, one may wonder whether $P_{Y1}(\Theta)$ could instead be interpreted as "the probability of *at most* (instead of *at least*) an α degree of satisfaction to the condition Θ ". In either case, it is challenging

to understand what "the probability of at least/at most an α degree of satisfaction to the condition Θ " truly means; (iii) In $P_K(\Theta)$, the supremum operation was again used. Consequently, would $P_K(\Theta)(w)$ simply correspond to "degree of truth" or in fact to "the maximum degree of truth" for the associated proposition? (iv) In $P_{Y2}(\Theta)$, an FST user may legitimately be worried about the suitability of the use of the "min" operator for one's application domain (e.g., could it be more appropriate to take the average of $P_K(\Theta)(w)$ and $P_{Y1}(\Theta)(w)$ instead?); (v) Finally, in $P_{HS}(\Theta)$, the practical implication of and justification for using the α -cuts $\Theta_{\alpha/2}$ and $\Theta_{1-\alpha/2}$ to calculate the probability remains opaque (e.g., why not use the α -cuts $\Theta_{\alpha/3}$ and $\Theta_{1-\alpha/3}$ instead?). In contrast, as illustrated in the examples in Section 3.1, our proposed extension principle allows for the derivation and determination of the probability of fuzzy events through step-by-step operations that are well understood by classically trained engineers.

The FST literature on fuzzy correlation is quite vast. Thus, we will limit our review here to a few highly cited papers: In an early work, Liu and Kao (2002) pointed out the deficiencies of previous approaches (e.g., the resulting coefficient being in the range $[0,1]$ only, or the coefficient being a crisp number hence lacking fuzziness). Consequently, Liu and Kao (2002) and Hong (2006) developed new correlation coefficient measures which yielded fuzzy numbers (but in turn having the problem of possibly exceeding the range $[-1,1]$). Hung (2001) developed a correlation measure for intuitionistic fuzzy sets via a statistical approach, which was further extended in Szmidt and Kacprzyk (2010). Liu, Shen, Mu, Chen, and Chen (2016) proposed fuzzy correlation measure based on concepts such as deviation and variance. Thao, Ali, and Smarandache (2019) developed an approach which could be viewed as an enhancement of that of Liu et al. (2016). More recent highly cited works include Ejegwa, Wen, Feng, Zhang, and Tang (2022), Ganie, Singh, and Bhatia (2020) and Lin, Huang, Chen, Fujita, and Wang (2021), developing correlation measures for picture fuzzy sets and Pythagorean fuzzy sets. On the whole, these works have brought richness to the types of fuzzy correlation measures that one can choose from for his/her applications. However, the measures proposed in these works are predominantly heuristically postulated rather than built on fundamental principles. Additionally, these works provide no mechanism for a user to specify interactivity among his/her fuzzy data of interest. In contrast, as illustrated in Section 3.2, our proposed extension principle allows for the development (rather than heuristic postulation) of fuzzy-valued correlation coefficients that can incorporate user-specified interactivity among the fuzzy data. Furthermore, our methodology is quite general and should be applicable to deriving other fuzzy-valued statistical measures.

5. Conclusion

Zadeh's Extension Principle (ZEP) has been indispensable for transforming various mathematical functions for classical crisp sets into the corresponding functions for fuzzy sets. However, some of the key elements in ZEP are arguably non-trivial for FST users who do not possess deep FST expertise to comprehend and interpret. Furthermore, directly applying ZEP to fundamental tasks, such as extending classical arithmetic to the fuzzy setting, can yield results that are too imprecise for real-life applications.

In this article, we propose an alternative FST extension principle composed of concrete and comprehensible elements for FST users. Consequently, an FST user employing the proposed principle will not only obtain a solution to his problem but also concretely understand the meaning of each step leading to the solution. Moreover, the proposed extension principle allows FST users to construct fuzzified functions that are adaptive to assumptions in their problem domains. Finally, we demonstrate the versatility of the proposed principle as a unified method for completing various non-trivial tasks, such as calculating probabilities of fuzzy events and constructing fuzzy-valued statistical measures, all in a user-interpretable manner.

There are limitations in the present work that nonetheless represent opportunities for future investigations:

- (i) We have exclusively used the K -intervals representation to decompose given fuzzy numbers. While it has clear advantages over the α -cut based approaches, the K -intervals representation is not necessarily the only way to perform decomposition on a fuzzy number. Thus, it is of interests to identify methods for fuzzy number decomposition in a more problem-adaptive way.
- (ii) As part of the requirements for using the proposed principle, an FST user must supply matching patterns, and the need to prescribe matching patterns represents additional work for the user. However, we believe that extra effort of this type can offer tremendous payoffs. This situation is akin to fundamental problems in other mathematical domains, such as partial differential equation (PDE) theory: A given PDE modeling a particular physical situation might not be solvable, even if the necessary initial and boundary conditions are provided. A fruitful approach to resolving ill-posed PDE problems is regularization, which involves supplying additional internal structures to augment the original physical model described by the PDE (see [Ngan & Truskinovsky, 2002](#) for an example). When done appropriately, such augmentations can successfully resolve an initially ill-posed PDE. In the present situation, prescribing matching patterns can be seen as a concrete way to supply necessary “internal structures” so that the fuzzification of a given mathematical function can be obtained in a problem-adaptive manner.
- (iii) Last but not least, the matching patterns considered in the present work describe pairwise relations, e.g., for a pair of intervals or a pair of fuzzy numbers. For more precise modeling, there may be a need to consider higher-order matching patterns, such as a triplet

or quadruplet of entities being considered simultaneously. The feasibility of doing so in a user-interpretable manner requires further study.

The present framework can be extended in various ways. We are particularly interested in applying it to develop alternative approaches for joint possibility distributions, fuzzy logic, and higher-order fuzzy logic. Other research directions include extending the present framework to obtain concrete extension principles for type-2 fuzzy sets and other higher-order fuzzy sets. We are also interested in using the framework to broaden and deepen the developments of probability and statistical theories of fuzzy events and data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX A: Membership value of an element supported by multiple sources of evidence

Consider two fuzzy numbers, A_1 and A_2 , each defined by the following triangular membership function (refer to Fig. A1a for the membership function and Fig. A1b for the corresponding K -intervals representation, with $K = 10000$):

$$\mu_{A_n}(x) = \begin{cases} (x_n + 4)/4 & \text{if } -4 < x_n \leq 0 \\ (4 - x_n)/4 & \text{if } 0 < x_n < 4 \\ 0 & \text{otherwise} \end{cases} \text{ with } n \in \{1, 2\} \quad (\text{A.1})$$

Further, suppose we are given a real-valued function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f(w, z) \equiv w \bullet z \text{ for any } w, z \in \mathbb{R} \quad (\text{A.2})$$

Our current problem is to extend f into its fuzzified version $\hat{f}$, thereby determining the fuzzy number $\hat{f}(A_1, A_2)$, which represents the product of A_1 and A_2 . This can be achieved by applying the FST extension principle for the $\mathbb{R}^2$ case (see [Definition 2.12](#) and the associated diagram in [Fig. 15](#)).

As the application of [Definition 2.12](#) has been detailed in [Section 3.1.2](#), we will simply present the key results for the current problem: Let us use the perfectly correlated matching pattern for the DM level and the uncorrelated matching pattern the mind-frame level:

$$\Pi_{A_1, A_2} \equiv \left\{ a_{A_1, A_2}^{k_1, k_2} \mid k_1, k_2 \in \{1, \dots, K\} \right\} \quad (\text{A.3})$$

$$\text{with } a_{A_1, A_2}^{k_1, k_2} = \begin{cases} 1 & \text{if } k_1 = k_2 \\ 0 & \text{otherwise} \end{cases} \text{ for } k_1, k_2 \in \{1, \dots, K\}$$

$$\text{and } \Pi_{A_1^{k_1}, A_2^{k_2}} \equiv \left\{ a_{A_1^{k_1}, A_2^{k_2}}^{m_1, m_2} \mid m_1, m_2 \in \{1, \dots, M\} \right\} \quad (\text{A.4})$$

$$\text{with } a_{A_1^{k_1}, A_2^{k_2}}^{m_1, m_2} = 1 \text{ for all } m_1, m_2 \in \{1, \dots, M\}$$

By performing the calculations required in parts (a) and (b) of [Definition 2.12](#), $C_{A_1, A_2}^{f,1}$ and $C_{A_1, A_2}^{f,2}$ (i.e., Eqs. (2.39) and (2.40)) can be determined. In fact, since the perfectly correlated matching pattern for the DM level has been employed, $C_{A_1, A_2}^{f,1} = C_{A_1, A_2}^{f,2} = \{\tilde{f}_k \mid k \in \{1, \dots, K\}\}$, and these resultant intervals $\tilde{f}_k$'s are displayed in Fig. A1c. Next, performing the calculations required in part (c) of [Definition 2.12](#) yields Fig. A1d, the membership function of the fuzzy number $\hat{f}(A_1, A_2)$. Alternatively, if one uses ZEP to determine the product of A_1 and A_2 , the resultant membership function would be:

$$\mu_{\hat{f}(A_1, A_2)}(y) \equiv \begin{cases} \sup_{(x_1, x_2) \in f^{-1}(y)} \{\min\{\mu_{A_1}(x_1), \mu_{A_2}(x_2)\}\} & \text{when } f^{-1}(y) \text{ is not empty} \\ 0 & \text{otherwise} \end{cases} \quad (\text{A.5.1})$$

$$\text{where } f^{-1}(y) = \{(x_1, x_2) \in \mathbb{R}^2 | x_1 \bullet x_2 = y\} \quad (\text{A.5.2})$$

leading us to Fig. A2.

Let's examine a typical case, where $y = 9$. There are infinitely many solutions to $f^{-1}(9)$ (e.g., $(x_1 = 2.5, x_2 = 3.6)$, $(x_1 = 2.25, x_2 = 4)$). Among these, both $(x_1 = 3, x_2 = 3)$ and $(x_1 = -3, x_2 = -3)$ yield $\min\{\mu_{A_1}(x_1), \mu_{A_2}(x_2)\} = 0.25$, while all other solutions result in values less than 0.25. Thus, based on Eq. (A.5.1), which takes the supremum of these values, $\mu_{\hat{f}(A_1, A_2)}(9) = 0.25$ (as depicted in Fig. A2), representing the degree of belongingness of 9 to the fuzzy number $\hat{f}(A_1, A_2)$ according to ZEP. Apparently, this result does not account for multiple sources of evidence (e.g., $(x_1 = -3, x_2 = -3)$ and $(x_1 = 3, x_2 = 3)$) supporting the membership of 9 to $\hat{f}(A_1, A_2)$.

Using our proposed extension principle, the K -intervals representations of A_1 and A_2 are shown in Fig. A1b. Recall that we set $K = 10,000$ (i.e., there are a total of 10,000 DMs). Then:

- (i) From Fig. A1b, we see that the value -3 is contained in intervals A_1^k and A_2^k for $k \in \{7501, 7502, \dots, 10000\}$. This means that DM k , for $k \in \{7501, 7502, \dots, 10000\}$, regard the value -3 as belonging to A_1 and A_2 . For these DMs, the value $(-3) \times (-3) = 9$ will also be contained within their corresponding intervals $\bar{f}_k$ (for $k \in \{7501, 7502, \dots, 10000\}$; see Fig. A1c) for calculating the product of A_1 and A_2 .
- (ii) Similarly, from Fig. A1b, we see that the value 3 is contained in intervals A_1^k and A_2^k for $k \in \{1, 2, \dots, 2500\}$. This means that DM k , for $k \in \{1, 2, \dots, 2500\}$, regard the value 3 as belonging to A_1 and A_2 . For these DMs, the value $3 \times 3 = 9$ will also be contained within their corresponding intervals $\bar{f}_k$ (for $k \in \{1, 2, \dots, 2500\}$; see Fig. A1c) for calculating the product of A_1 and A_2 .
- (iii) As a result, out of the 10,000 DMs, the fraction who regard the value 9 as belonging to $A_1 \times A_2$ is $(2500 + 2500)/10000 = 0.5$. This coincides with the result indicated in Fig. A1d, with $\mu_{\hat{f}(A_1, A_2)}(9) = 0.5$.

This example demonstrates that the proposed extension principle can accommodate the intuition that multiple sources of evidence for a hypothesis may increase its degree of belongingness to the fuzzy set of concern.

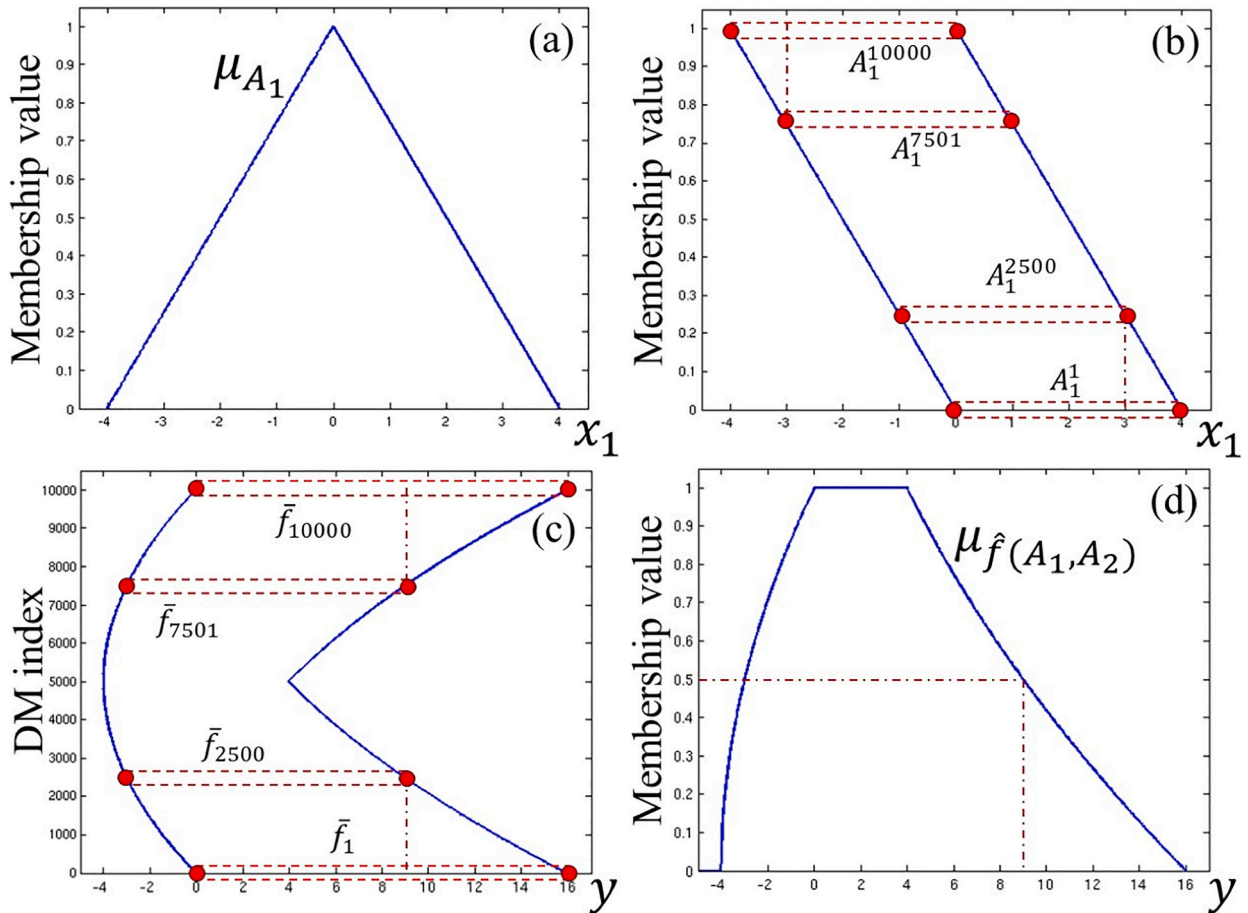


Fig. A1. Calculation of the product of A_1 and A_2 . (a) The membership function of the fuzzy number A_1 . According to Eq. (A.1), A_2 has the same membership function. (b) The K -intervals representation of A_1 , with $K = 10,000$. Since A_2 shares the same membership function, its K -intervals representation is identical to A_1 . Note that intervals A_1^1 to A_1^{2500} contain the value 3, while intervals A_1^{7501} to A_1^{10000} contains the value -3 . (c) The resultant intervals for calculating $\mu_{\hat{f}(A_1, A_2)}$. Interval $\bar{f}_k$ represents the range of values regarded by DM k as belonging to the product of A_1 and A_2 . Note that intervals $\bar{f}_1$ to $\bar{f}_{2500}$ and $\bar{f}_{7501}$ to $\bar{f}_{10000}$ contains the value 9. (d) The membership function $\mu_{\hat{f}(A_1, A_2)}$ of the resultant fuzzy number $\hat{f}(A_1, A_2)$, the product of A_1 and A_2 .

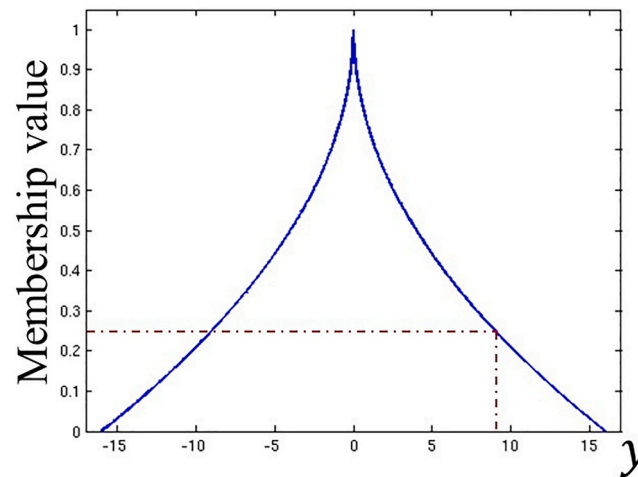


Fig. A2. Membership function for the product of A_1 and A_2 according to ZEP.

Data availability

No data was used for the research described in the article.

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